



华沿机器人
HUAYAN ROBOTICS

广东华沿机器人焊接工艺包

控制通信协议

(Elfin 系列机器人通用)



修订记录

文件编号	HR-TCP-V3.x		
文件状态	[]草稿 [] 正式发布 [√]正在修改		
当前版本	V3.15.0		
拟 制	广东华沿机器人应用软件组	日期	2025 年 4 月 30 日
审 核	广东华沿机器人系统测试组	日期	2025 年 4 月 30 日
批 准	广东华沿机器人	日期	2025 年 4 月 30 日
发布日期	2025 年 4 月 30 日		
生效日期	2025 年 4 月 30 日		

A - 增加 M - 修订 D - 删除

版本编号	版本日期	焊接工艺包版本	支持的控制器版本	更新说明
V3.8.0	2025.4.30	3.8.x	V6.4.16C 及以上	M: 初始版本
V3.10	2025.7.4	3.10.x	V6.5 及以上	M: 修改摆动参数 1 shape 的取值类型发送变化，由 string 变为 int，0 表示正弦摆，2 表示三角摆； 2 增加 centralUplift, 表示中央隆起的高度，单位是毫米。 3 删除了预留字段 centreRise

V3.15	2025.11.30	3.15.x	V6.4.16C/6.5.19b 及以上	1 将 DeviceOn 焊机的配置拆分新接口：DeviceConfig, 焊机正常连接可以单独修改相关配置参数 2 起弧补充切割字段描述 3 新增切割弧压跟踪模板 4 摆幅、频率 类型为 double；两端停留时间类型为 double； 5 摆动 新增 suspendType 字段, 类型 int, 停留类型：0 停止, 1 前进 6 更新圆弧 movecType 的取值范围 7 更新接触寻位章节, 补充场景报文
V3.15	2026.03.07	3.15.x	V6.4.16C/6.5.19b 及以上	A: 新增实时焊接参数修改接口 新增原地摆动接口

V3.17	2026.04.30	3.17.x	V6.4.16C/6.5.19b 及以上	M: 电弧跟踪接口, 新增 tcpMode 参数
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第一章 概述

1.1 编写目的

为使用广东华沿机器人焊接工艺包通信协议进行二次开发的人员提供接口使用说明。

1.2 使用须知

1. 广东华沿机器人焊接工艺包通信协议接口主要提供广东华沿机器人接入焊接应用通信协议标准,使用人员依照此协议内容通过 TCP/IP 发送指定消息到广东华沿机器人焊接工艺包系统(以下简称焊接系统),焊接系统依据不同的消息内容分别处理,并将处理结果返回发送方。
2. 焊接系统每一次只会处理一个完整的消息,如一次发送多条消息或者发送的消息不完整,焊接系统会返回命令无效的结果。
3. 表格中未列出的字段均为前端使用,不对后端产生影响,可使用默认值。
4. 本文档的接口只支持下表所列出的焊机品牌使用

表 1-1 华沿机器人支持的焊机列表

支持焊机品牌	对接协议
奥太焊机	Modbus TCP
麦格米特	EtherNet/IP
和宗	EtherNet/IP
宾采尔	Modbus TCP
肯倍	Modbus TCP
威尔泰克	EtherNet/IP

海宝	RS485
克鲁斯	EtherNet/IP
默克莱	EtherNet/IP
福尼斯	Modbus TCP
洛驰	Modbus TCP
林肯	EtherNet/IP
伊达新	EtherNet/IP
米勒	Modbus TCP
	EtherNet/IP
伊萨	EtherNet/IP
松下	EtherNet/IP
瑞凌	EtherNet/IP
所有品牌	IO

1.3.数据类型

表 1-2 数据类型

数据类型	长度	说明
string	-	字符串

int	-	整型
double	-	浮点型
array	-	数组
object	-	对象

第二章 消息格式

2.1.通信协议格式

消息示例：

```
####0000006a{"function":"continueTask","parameter":{"continueType":0,"backtrackingDistance":5,"backtrackingSpeed":50}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "continueTask",
  "parameter": {
    "continueType": 0,
    "backtrackingDistance": 5,
    "backtrackingSpeed": 50
  }
}
```

完整消息（包括指令和应答）都是 ASCII 码格式（字符串形式）。

表 2-1 消息格式说明

标识	说明
####	帧头
00000000	消息体长度，以 8 字节十六进制表示
{...}	消息体内容，为压缩 JSON 字符串
\$\$\$\$	帧尾

2.2.返回格式

2.2.1.成功返回

```
{"code":0,"error":""}
```

2.2.2.失败返回

```
{"code":102101,"error":"error message"}
```

2.2.3.参数说明

表 2-2 返回消息说明

参数	名称	数据类型	内容
code	状态码	int	0 为成功，其他为失败
error	错误消息	string	空或错误消息内容

2.3. 消息队列说明

采用了 ZeroMQ 轻量级的消息框架，使用了发布-订阅模式，推送消息状态，上位机需要实现客户端的订阅功能。具体参考：<https://zeromq.org/get-started/> 获取不同开发语言的样例。

- StatusMessage 消息固定标识
- taskStatus —— 任务状态，0：表示空闲，1：表示正在执行，2：表示继续，3：表示暂停，4：表示正在停止
- taskID —— 正在执行的任务 ID
- actionID —— 正在执行的动作（action）ID，要求固定 18 位，不重复（建议生成方式参考现在代码生成的方式，格式为：时间戳 + 5 位补零的计数器）

以 javascript 语言为例：

```
const actionID = timestamp.toString() + String(counter).padStart(5, '0');
```

- weldErrorCode —— 焊机错误状态码，具体看焊机的文档，0：表示焊机正常
- taskStatus - 任务状态，数字类型
- taskType - 任务类型，数字类型
- reservationStatus - 预约状态，数字类型
- deviceStatus - 设备状态列表，数组类型，每个设备包含如下：
- name - 设备名称，字符串类型
- type - 设备类型，数字类型
- model - 设备型号，字符串类型
- protocol - 协议类型，数字类型
- state - 设备状态，数字类型
- ip - IP 地址，字符串类型
- weldMode - 焊接模式，数字类型（针对焊机）
- teachMode - 示教模式，数字类型（针对焊机）
- realtimeWireSpeed - 实时送丝速度，数字类型（针对焊机）

- cartridge - 切割信息，对象类型（针对 Hypertherm 设备）
- port - 端口号，数字类型（针对相机设备）
- laserWeldState - 激光焊接状态，数字类型（针对 Industry 设备）

周期发送消息：（包含所有状态字段，周期 100ms，端口：30601），格式：

```
StatusMessage {  
  "taskID": "...",  
  "actionID": "...",  
  "actionResult": "...",  
  "taskStatus": "...",  
  "taskType": "...",  
  "reservationStatus": "...",  
  "deviceStatus": [  
    {  
      "name": "...",  
      "type": "...",  
      "model": "...",  
      "protocol": "...",  
      "state": "...",  
      "ip": "...",  
      "weldMode": "...",  
      "teachMode": "...",  
      "weldErrorCode": "...",  
      "realtimeWireSpeed": "...",  
    },  
    "...",  
  ],  
  "laserWeldState": "..."  
}
```

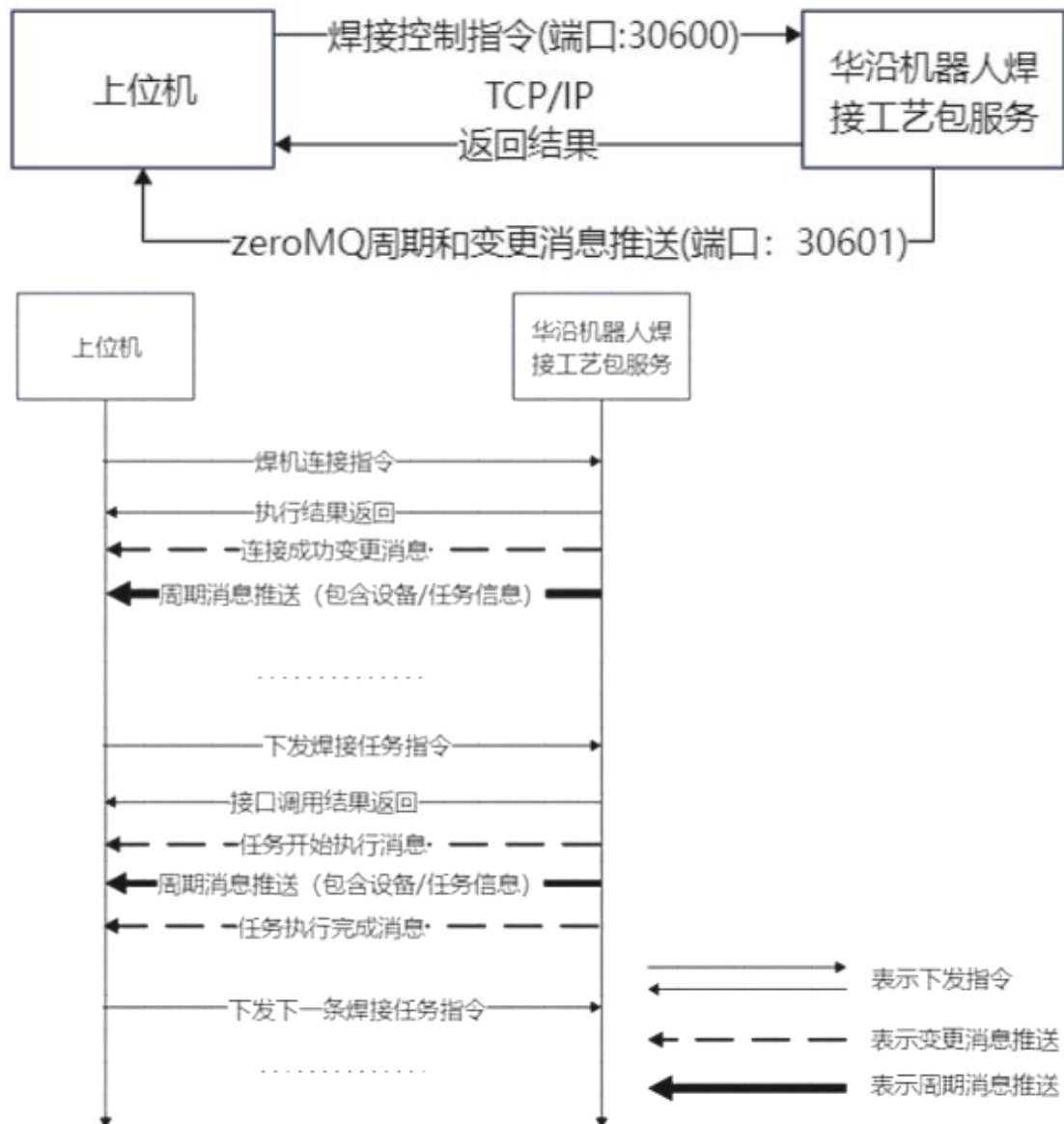
第三章 服务端

协议采用标准的 C/S 架构，其中广东华沿机器人提供接收并处理焊接控制消息的服务端，整个通信过程采用 TCP/IP 方式进行。

3.1 连接方式

3.1.1 作为服务器

广东华沿机器人焊接系统作为通信服务端，启动时，会不断监听指定端口，等待上位机客户端的连接。



3.1.2 连接示例

3.1.2.1 同一台工控机进行通信

1. 控制器作为服务端，客户软件作为客户端，以 NetAssist.exe 为例。NetAssist 作为客户端，IP 地址为本机地址，端口 30600：



图 3-1 连接示例

3.1.2.2 工控机与笔记本电脑通信

1. 使用网线将工控机和笔记本电脑连接；
2. 修改工控机和笔记本电脑 IP 地址，使其在同一个域。
 - a. 进入管理-网络，修改工控机 IP:

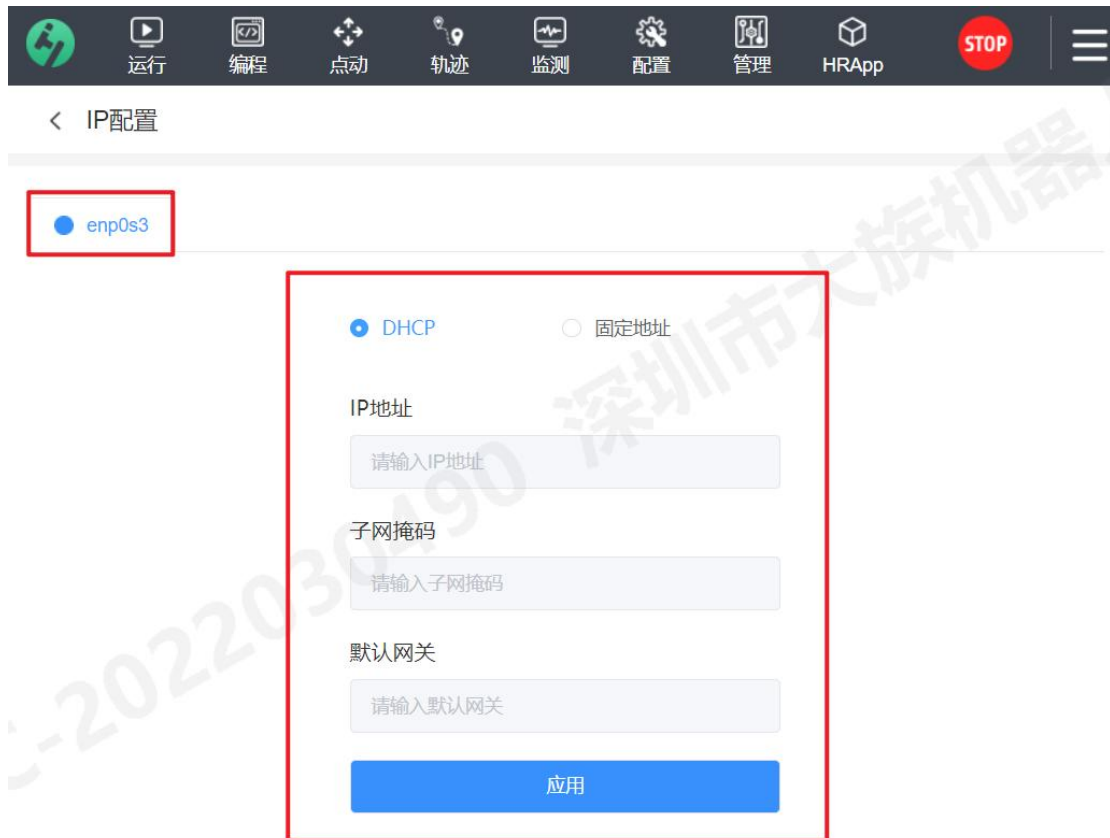


图 3-2 修改工控机 IP

b. 修改笔记本 IP:

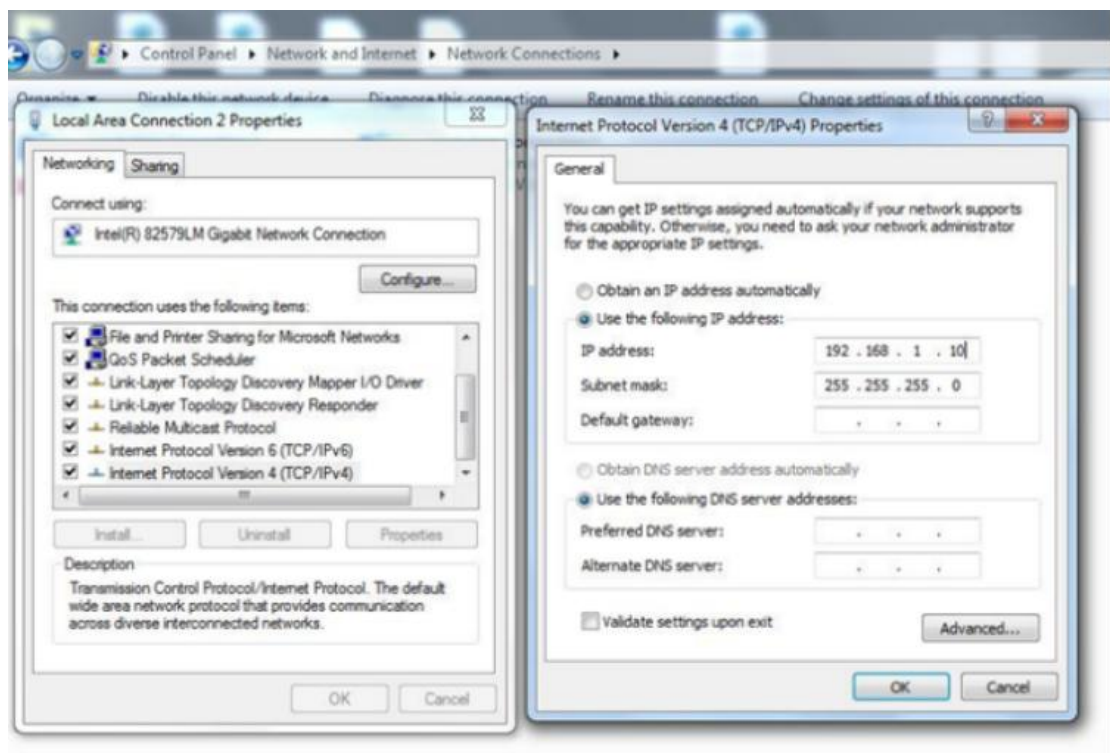


图 3-3 修改笔记本 IP

3. 控制器作为服务端，客户软件作为客户端，以 NetAssist.exe 为例。NetAssist 作为客户端，IP 地址为工控机地址，端口 30600：



图 3-4 连接示例

3.2 服务端口表

服务端口	说明
30600	焊接控制指令服务端口
30601	焊接控制消息队列端口

3.3 机器人控制电箱 IO 口表

下表是根据标准电箱提供的列表，具体以实际电箱型号为准。

表 3-1 华沿机器人控制电箱 IO 口列表

输入端口	输出端口
CI0/DI0	CO0/DO0
CI1/DI1	CO1/DO1
CI2/DI2	CO2/DO2
CI3/DI3	CO3/DO3
CI4/DI4	CO4/DO4
CI5/DI5	CO5/DO5
CI6/DI6	CO6/DO6
CI7/DI7	CO7/DO7
	AO0/AO1 (输出:0~10V)

第四章 通信协议接口

4.1 设备管理

4.1.1 DeviceOn

4.1.1.1 描述:连接设备

发送消息

```
####00000068{"function":"DeviceOn","parameter":{"name":"IO","protocol":0,"ip":"","port":null,"model":"IO","type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function":"DeviceOn",
  "parameter":{
    "name":"IO",
    "protocol":0,
    "ip":"",
    "port":null,
    "model":"IO",
    "type":0
  }
}
```

表 4-1-1 连接设备-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: DeviceOn
parameter	参数	object	参数对象
name	设备名称	string	取值参考: 5.1 设备名称/协议
protocol	协议类型	int	取值参考: 5.1 设备名称/协议
ip	设备 IP	string	焊机 IP, 如果无, 填写: "127.0.0.1"
port	设备端口	int	如果无, 填写: null
model	设备型号	arr	取值参考: 5.2 设备型号
type	设备类型	int	0: 焊机; 1: 外设激光 2: 外设相机

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.2 DeviceOff

4.1.2.1 描述:关闭连接

发送消息：

```
####0000002F{"function":"DeviceOff","parameter":{"type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "DeviceOff",
  "parameter": {
    "type": 0,
  }
}
```

表 4-1-2 关闭连接-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：DeviceOff
parameter	参数	object	见下表
type	设备类型	int	0：焊机，1：激光，2：相机

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.3 ClearError

4.1.3.1 描述:清错

发送消息：

```
####00000028{"function":"clearError","parameter":{}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "clearError",
  "parameter": {}
}
```

表 4-1-3 清错-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: clearError
parameter	参数	object	填写为 {}

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.4 ChangeMode

4.1.4.1 描述: 切换下发模式 (仅麦格米特)

发送消息:

```
####00000041{"function":"changeMode","parameter":{"enable":1,"unit":"m/min"}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "changeMode",
  "parameter": {
    "enable": 1,
    "unit": "m/min"
  }
}
```

表 4-1-4 切换效法模式消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: changeMode
parameter	参数	object	见下表
enable	是否启用	int	0: 不启用送丝速度 1: 启用送丝速度
unit	单位	string	"m/min": 米每分钟 "inch/min": 英寸每分钟

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.5 ArcSet

4.1.5.1 描述: 下发电流电压

发送消息:


```
####000001a7{"function":"arcSet","parameter":{"type":"arcSet","arcSetMode":0,"arcSetCurrent":100,"arcSetVoltage":0,"arcSetAirPressure":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetSlope":0,"arcSetInductance":0,"arcSetInterpulse":0,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetBurnBackTime":0,"arcSetArcControl":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "arcSet",
  "parameter": {
    "type": "arcSet",
    "arcSetMode": 0,
    "arcSetCurrent": 100,
    "arcSetVoltage": 0,
    "arcSetAirPressure": 0,
    "arcSetPowerCorrection": 20,
    "arcSetWireSpeed": 0,
    "arcSetJobNumber": 1,
    "arcSetSlope": 0,
    "arcSetInductance": 0,
    "arcSetSwingLength": 0,
    "arcSetSwingFrequency": 0,
    "arcSetScanSpeed": 200,
    "arcSetScanLength": 2,
    "arcSetSpduty": 50,
    "arcSetSpfreq": 2000,
    "arcSetBurnBackTime": 0,
    "arcSetArcControl": 0,
  }
}
```

表 4-1-5 下发电流电压-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：ArcSet
parameter	参数	object	见下表
type	是否启用	string	固定值：arcSet

arcSetMode	焊接模式	int	见 5.3 焊接模式
arcSetCurrent	电流值	double	焊接/切割电流
arcSetVoltage	电压值	double	分别模式生效，其他模式，填写：0
arcSetPowerCorrection	功率校正	double	一元化模式生效，其他模式，填写：0
arcSetWireSpeed	送丝速度	double	非麦格米特，填写：0
arcSetJobNumber	焊机 job 号	int	非 job 模式，填写：0
arcSetBurnBackTime	回烧时间	int	依据焊机实际情况，无则填写：0
arcSetInductance	电感	int	依据焊机实际情况，无则填写：0
arcSetSlope	坡度	int	依据焊机实际情况，无则填写：0
arcSetAirPressure	切割气压	int	依据焊机实际情况，无则填写：0
arcSetSwingLength	摆动宽度	int	依据焊机实际情况
arcSetSwingFrequency	摆动频率	double	依据焊机实际情况
arcSetScanSpeed	扫描速度	double	依据焊机实际情况
arcSetScanLength	扫描宽度	int	依据焊机实际情况
arcSetSpduty	占空比	int	依据焊机实际情况，无则填写：0
arcSetSpfreq	激光频率	int	依据焊机实际情况，无则填写：0

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.6 ArcOn

4.1.6.1 描述：起弧

发送消息：

```
####000000ef{"function":"arcOn","parameter":{"type":"arcOn","arcOnMode":0,"arcOnCurrent":100,"arcOnVoltage":0,"arcOnAirPressure":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnSlope":0,"arcOnInductance":0,"arcOnArcControl":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "arcOn",
  "parameter": {
    "type": "arcOn",
    "arcOnMode": 0,
    "arcOnCurrent": 100,
    "arcOnVoltage": 0,
    "arcOnAirPressure": 0,
    "arcOnPowerCorrection": 20,
    "arcOnWireSpeed": 0,
    "arcOnJobNumber": 1,
    "arcOnSlope": 0,
    "arcOnInductance": 0,
    "arcOnArcControl": 0
  }
}
```

表 4-1-6 起弧-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：arcOn
parameter	参数	object	见下表
type	是否启用	string	固定值：arcOn
arcOnMode	下发模式	int	见 5.3 焊接模式
arcOnCurrent	电流值	double	焊接/切割电流
arcOnVoltage	电压值	double	分别模式生效，其他模式，填写：0
arcOnPowerCorrection	功率校正	double	一元化模式生效，其他模式，填写： 0
arcOnWireSpeed	送丝速度	double	非麦格米特，填写：0
arcOnJobNumber	job 号	int	非 job 模式，填写：0
arcOnSlope	设置斜率	double	依据焊机实际情况，无则填写：0
arcOnInductance	设置电感	int	依据焊机实际情况，无则填写：0
arcOnAirPressure	切割气压	int	依据焊机实际情况，无则填写：0
arcOnArcControl	电弧控制	int	依据焊机实际情况，无则填写：0

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.7 ArcOff

4.1.7.1 描述：收弧

发送消息：

```
####000000e6{"function":"arcOff","parameter":{"type":"arcOff","arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffAirPressure":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffSlope":0,"arcOffInductance":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "arcOff",
  "parameter": {
    "type": "arcOff",
    "arcOffMode": 0,
    "arcOffCurrent": 100,
    "arcOffVoltage": 0,
    "arcOffAirPressure": 0,
    "arcOffPowerCorrection": 20,
    "arcOffWireSpeed": 0,
    "arcOffJobNumber": 1,
    "arcOffSlope": 0,
    "arcOffInductance": 0,
  }
}
```

表 4-1-7 收弧消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：arcOff
parameter	参数	object	见下表

type	是否启用	string	固定值: arcOff
arcOffMode	下发模式	int	见 5.3 焊接模式
arcOffCurrent	电流值	double	焊接电流
arcOffVoltage	电压值	double	分别模式生效, 其他模式, 填写: 0
arcOffPowerCorrection	功率校正	double	一元化模式生效, 其他模式, 填写: 0
arcOffWireSpeed	送丝速度	double	非麦格米特, 填写: 0
arcOffJobNumber	job 号	int	非 job 模式, 填写: 0
arcOffSlope	收弧电感	int	依据焊接实际情况, 无则填写: 0
arcOffInductance	收弧斜率	int	依据焊接实际情况, 无则填写: 0

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.8 startFlow

4.1.8.1 描述: 送气

发送消息:

```
####00000039{"function":"startFlow","parameter":{"type":"startFlow"}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "startFlow",
  "parameter": {
    "type": "startFlow"
  }
}
```

表 4-1-8 送气-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: startFlow
parameter	参数	object	见下表
type	类型	string	固定值: startFlow

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.9 StopFlow

4.1.9.1 描述: 停止送气

发送消息:

```
####00000037{"function":"stopFlow","parameter":{"type":"stopFlow"}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "stopFlow",
  "parameter": {
    "type": "stopFlow"
  }
}
```

表 4-1-9 停止送气消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: stopFlow
parameter	参数	object	见下表
type	类型	string	固定值: stopFlow

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.10 StartPushWire

4.1.10.1 描述: 送丝

发送消息:


```
####00000041{"function":"startPushWire","parameter":{"type":"startPushWire"}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "startPushWire",
  "parameter": {
    "type": "startPushWire"
  }
}
```

表 4-1-10 送丝-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：startPushWire
parameter	参数	object	见下表
type	类型	string	固定值：startPushWire

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.11 StopPushWire

4.1.11.1 描述：停止送丝

发送消息：

```
####0000003f{"function":"stopPushWire","parameter":{"type":"stopPushWire"}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "stopPushWire",
  "parameter": {
    "type": "stopPushWire"
  }
}
```

表 4-1-11 停止送丝-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: stopPushWire
parameter	参数	object	见下表
type	类型	string	固定值: stopPushWire

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.12 StartPullWire

4.1.12.1 描述: 退丝

发送消息：

```
####00000041{"function":"startPullWire","parameter":{"type":"startPullWire"}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "startPullWire",
  "parameter": {
    "type": "startPullWire"
  }
}
```

表 4-1-12 退丝-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：startPullWire
parameter	参数	object	见下表
type	类型	string	固定值：startPullWire

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.13 StopPullWire

4.1.13.1 描述：停止退丝

发送消息：

```
####0000003f{"function":"stopPullWire","parameter":{"type":"stopPullWire"}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "stopPullWire",
  "parameter": {
    "type": "stopPullWire"
  }
}
```

表 4-1-13 停止退丝-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：stopPullWire
parameter	参数	object	见下表
type	类型	string	固定值：stopPullWire

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.14 HandleConfig

4.1.14.1 描述：设置手柄配置

发送消息：

```
####00000253{"function":"handleConfig","parameter":{"buttonsEnable":1,"buttonsConfig":
[{"buttonType":1,"longPressFunction":"pushAddJointMotion","shortPressFunction":"pushA
ddLinearMotion"},{"buttonType":1,"longPressFunction":"pushDragSwitch","shortPressFunc
tion":"pushAddCircleMotion"},{"buttonType":0,"longPressFunction":"","shortPressFunction":
"pushAddLinearWeld"},{"buttonType":0,"longPressFunction":"","shortPressFunction":"push
AddCircularWeld"},{"buttonType":0,"longPressFunction":"","shortPressFunction":"pushAdd
WeaveWeld"},{"buttonType":0,"longPressFunction":"","shortPressFunction":"pushAddArcO
n"}]]}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "handleConfig",
  "parameter": {
    "buttonsEnable": 1,
    "buttonsConfig": [
      {
        "buttonType": 1,
        "longPressFunction": "pushAddJointMotion",
        "shortPressFunction": "pushAddLinearMotion"
      },
      {
        "buttonType": 1,
        "longPressFunction": "pushDragSwitch",
        "shortPressFunction": "pushAddCircleMotion"
      },
      {
        "buttonType": 0,
        "longPressFunction": "",
        "shortPressFunction": "pushAddLinearWeld"
      },
      {
        "buttonType": 0,
        "longPressFunction": "",
        "shortPressFunction": "pushAddCircularWeld"
      },
      {
        "buttonType": 0,
        "longPressFunction": "",
        "shortPressFunction": "pushAddWeaveWeld"
      },
      {
        "buttonType": 0,
        "longPressFunction": "",
        "shortPressFunction": "pushAddArcOn"
      }
    ]
  }
}
```

```

    }
  ]
}
}

```

表 4-1-14 设置手柄配置-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	handleConfig
parameter	参数	object	见下表
buttonsConfig	按钮配置	array	按钮配置数组
buttonType	按钮类型	number	按钮类型值（1:支持长按和短按，0:短按）
longPressFunction	长按功能	string	长按执行的功能 pushAddJointMotion: 关节运动 pushAddLinearMotion: 直线运动 pushDragSwitch: 拖拽开关 pushAddCircleMotion: 圆弧运动 pushAddLinearWeld: 直焊模版 pushAddCircularWeld: 圆焊模版 pushAddWeaveWeld: 摆焊模版 pushAddArcOn: 起弧
shortPressFunction	短按功能	string	短按执行的功能

buttonsEnable	buttonsEnable	number	1: 启用手柄开关, 0:禁用手柄开关
---------------	---------------	--------	---------------------

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.15 SetAotaiMode

4.1.15.1 描述: 切换奥泰焊机模式

发送消息:

```
####00000037{"function":"setAotaiMode","parameter":{"aotaiMode":1}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "setAotaiMode",
  "parameter": {
    "aotaiMode": 1
  }
}
```

表 4-1-15 切换奥泰焊机模式-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: setAotaiMode
parameter	参数	object	见下表

aotaiMode	类型	int	0:电压 1:弧长矫正
-----------	----	-----	----------------

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.16 SetForceControlState

4.1.16.1 描述：切换力控状态

发送消息：

```
####0000002C{"function":"setForceControlState","parameter":"0"}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "setForceControlState",
  "parameter": 0
}
```

表 4-1-16 切换力控状态-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：setForceControlState
parameter	参数	int	0:关闭力控，1:开启力控

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.17 SetBoxDO

4.1.17.1 描述：设置 DO 信号

发送消息：

```
####0000003a{"function":"setBoxDO","parameter":{"state":false,"DO":1}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "setBoxDO",
  "parameter": {
    "state": false,
    "DO": 1
  }
}
```

表 4-1-17 设置 DO 信号-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：setBoxDO
parameter	参数	object	见下表
state	状态	int	false:关闭，true:开启

D0	输出信号 D0	int	固定值：1
----	---------	-----	-------

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.18 SetForceControlMode

4.1.18.1 描述：切换力控模式

发送消息：

```
####00000031{"function":"setForceControlMode","parameter":""}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "setForceControlMode",
  "parameter": ""
}
```

表 4-1-18-切换力控模式消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值： setForceControlMode
parameter	参数		

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.19 SetFroniusTeachMode

4.1.19.1 描述：切换 Fronius 示教模式

发送消息：

```
####0000003e{"function":"setFroniusTeachMode","parameter":{"teachMode":1}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "setFroniusTeachMode",
  "parameter": {
    "teachMode": 1
  }
}
```

表 4-1-19 切换 Fronius 示教模式-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值： setFroniusTeachMode
parameter	参数	object	见下表
teachMode	示教模式	int	0:关闭, 1:开启

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.20 WeldProgram

4.1.20.1 描述：设置焊接程序（仅米勒 ModBus TCP/IP 模式）

发送消息：

```
#####000000a5{"function":"weldProgram","parameter":{"type":"weldProgram","isUseMemorySlot":1,"programNumber":1,"weldProcess":1,"weldWireType":0,"weldWireSize":0,"weldGasType":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "weldProgram",
  "parameter": {
    "type": "weldProgram",
    "isUseMemorySlot": 1,
    "programNumber": 1,
    "weldProcess": 1,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  }
}
```

表 4-1-20 设置焊接程序-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：weldProgram
parameter	参数	object	见下表

weldProgram	焊接程序	object	见 4.1.20（仅米勒 ModBus TCP/IP 模式）
isUseMemorySlot	使用存储槽	int	0: 使用, 1:不使用, -1:从设备管理界面直接控制焊机
programNumber	程序编号	int	1
weldProcess	焊接工艺	int	0:mig 模式, 1:pause 模式
weldWireType	焊丝类型	int	
weldWireSize	焊丝尺寸	int	
weldGasType	保护气类型	int	

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.21 Materials

4.1.21.1 描述：设置焊接材料

发送消息：

```
####00000114{"function":"materials","parameter":{"type":"materials","programNumber":0,"wireExtensions":1,"materials":["1","0","0"],"stableFusionDepth":0,"arcCharacteristic":2,"doublePulseFreAdj":8,"use":3,"internalFunctions":[{"No":60,"Value":3},{"No":61,"Value":4},{"No":62,"Value":5}]}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "materials",
  "parameter": {
    "type": "materials",
    "programNumber": 0,
    "wireExtensions": 1,
    "materials": [
      "1",
      "0",
      "0"
    ],
    "stableFusionDepth": 0,
    "arcCharacteristic": 2,
    "doublePulseFreAdj": 8,
    "use": 3,
    "internalFunctions": [
      {
        "No": 60,
        "Value": 3
      },
      {
        "No": 61,
        "Value": 4
      },
      {
        "No": 62,
        "Value": 5
      }
    ]
  }
}
```

表 4-1-21 设置焊接材料-消息字段说明

参数	名称	数据类型	内容
function	功能类型	string	materials
type	类型	string	materials

programNumber	程序编号	int	0
wireExtensions	干伸长	int	
stableFusionDepth	恒溶深控制	int	1: 启用, 0:禁用
arcCharacteristic	电弧特性	int	2
doublePulseFreAdj	双脉冲频率调整	int	8
use	用途	int	0:半自动, 1:自动, 2:OP1, 3:OP2
materials	材料列表	array	材料数组
materials[0]	材料 1	int	1
materials[1]	材料 2	int	0
materials[2]	材料 3	int	0
internalFunctions	内部功能列表	array	内部功能数组
internalFunctions[0].No	功能 1 编号	int	60
internalFunctions[0].Value	功能 1 值	int	3
internalFunctions[1].No	功能 2 编号	int	61
internalFunctions[1].Value	功能 2 值	int	4
internalFunctions[2].No	功能 3 编号	int	62

internalFunctions[2].Value	功能 3 值	int	5
--------------------------------	--------	-----	---

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.1.22 DeviceConfig

4.1.22.1 描述:焊机设备配置参数

发送消息

Aotai/Binzel:

```
####000000c3{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":500},"voltage":{"min":0,"max":50},"arcLength":{"min":-5,"max":5},"jobNumber":{"min":1,"max":100}}}}$$$$
```

Bingo:

```
####000000c3{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":500},"arcLength":{"min":-5,"max":5},"jobNumber":{"min":1,"max":100},"voltage":{"min":0,"max":50}}}}$$$$
```

Cloos:

```
####000000dc{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":1,"max":100},"wireSpeed":[{"min":2,"max":30,"unit":"m/min"}],"arcLength":{"min":-50,"max":50},"inductance":{"min":-50,"max":50}}}}$$$$
```

ESAB:

```
####000000ce{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":1,"max":100},"voltageCompensation":{"min":-32.75,"max":32.75},"wireSpeed":[{"min":0,"max":655.35,"unit":"m/min"}]}}}}$$$$
```

ESAB(Standard Pulse):

```
####000000b9{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":1,"max":1000},"voltage":{"min":8,"max":44},"wireSpeed":[{"min":0.8,"max":35,"unit":"m/min"}]}}}}$$$$
```


ESAB(Standard):

```
####0000006a{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":1,"max":1000}}}}$$$$
```

EWM:

```
####000000e9{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"wireSpeed":{"min":0,"max":25,"unit":"m/min"},"voltageCompensation":{"min":-9.9,"max":9.9},"jobNumber":{"min":0,"max":255},"programNumber":{"min":0,"max":15}}}}$$$$
```

Flama(ModbusTCP):

```
####000000d3{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":500},"voltage":{"min":0,"max":100},"voltageCompensation":{"min":-9.9,"max":9.9},"jobNumber":{"min":1,"max":9999}}}}$$$$
```

Flama(Ethernet/IP):

```
####000000b4{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":1000},"voltageCompensation":{"min":-10,"max":10},"jobNumber":{"min":1,"max":9999}}}}$$$$
```

Flama(RS485):

```
####00000104{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"wireSpeed":{"min":0,"max":30,"unit":"m/min"},"voltage":{"min":0,"max":100},"current":{"min":0,"max":1000},"voltageCompensation":{"min":-9.9,"max":9.9},"jobNumber":{"min":1,"max":9999}}}}$$$$
```

Fronius:

```
####00000068{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":0,"max":99}}}}$$$$
```

Huayuan:

```
####00000122{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":999},"voltage":{"min":-1000,"max":1000},"wireSpeed":{"min":0,"max":999,"unit":"m/min"},"arcLength":{"min":-1000,"max":1000},"jobNumber":{"min":1,"max":20},"programNumber":{"min":1,"max":17}}}}$$$$
```

Industry LaserWeld:

```
####00000068{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"voltage":{"min":0,"max":1500}}}}$$$$
```

IO:

```
####0000027b{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","ioConfig":{"AO0":{"type":3,"curvePoints":[{"key":0,"value":0}, {"key":1,"value":100}, {"key":10,"value":500}], "AO1":{"type":3,"curvePoints":[{"key":0,"value":-10}, {"key":2,"value":-5}, {"key":10,"value":10}], "functionMap":{"outPut":{"arcOn":{"key":"CO1","value":0}, "wirePush":{"key":"CO4","value":0}, "wirePull":{"key":"CO2","value":0}, "gasDetection":{"key":"","value":0}, "enableTouchSense":{"key":"","value":0}, "inPut":{"welderReady":{"key":"","value":0}, "arcDetection":{"key":"","value":0}, "touchSensed":{"key":"","value":0}, "robotEmergencyStop":{"key":"","value":0}}}}}}}}$$$$
```

IPG:

```
####000000f4{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"laserDuty":{"min":0,"max":100},"laserFrequency":{"min":0,"max":10000},"swingFrequency":{"min":0,"max":300},"swingLength":{"min":0,"max":5},"voltage":{"min":0,"max":1500}}}}$$$$
```

Kemppi:

```
####00000101{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":0,"max":99},"current":{"min":0,"max":1024},"wireSpeed":{"min":0.5,"max":25,"unit":"m/min"},"voltage":{"min":8,"max":46},"voltageCompensation":{"min":-10,"max":10}}}}$$$$
```

Lincoln:

```
####00000069{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":1,"max":100}}}}$$$$
```

Lorch:

```
####00000069{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":1,"max":100}}}}$$$$
```

Megmeet:

```
####00000127{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":500},"wireSpeed":{"min":0,"max":28,"unit":"m/min"},"min":0,"max":1102,"unit":"inch/min"},"voltageCompensation":{"min":-30,"max":30},"jobNumber":{"min":0,"max":100},"voltage":{"min":12,"max":38}}}}$$$$
```

Megmeet Laser:

```
####000000ba{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"voltage":{"min":0,"max":12000},"wireSpeed":{"min":0,"max":600,"unit":"cm/min"},"jobNumber":{"min":0,"max":99}}}}$$$$
```

Canaweld:

```
####000000ff{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"powerCorrection":{"min":0,"max":100},"voltageCompensation":{"min":-30,"max":30},"inductance":{"min":-100,"max":100},"jobNumber":{"min":1,"max":255},"interpulse":{"min":0,"max":100}}}}$$$$
```

Miller(ModbusTCP):

```
####000000dd{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"arcControl":{"min":0,"max":50},"arcLength":{"min":-10,"max":10},"voltage":{"min":10,"max":38},"wireSpeed":{"min":50,"max":780,"unit":"inch/min"}}}}$$$$
```

Miller(Ethernet/IP):

```
####000000df{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":500},"inductance":{"min":0,"max":99},"jobNumber":{"min":1,"max":100},"slope":{"min":0,"max":99},"voltage":{"min":1,"max":25}}}}$$$$
```

OTC:

```
####00000102{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":300},"unifyVoltage":{"min":-100,"max":100},"voltage":{"min":0,"max":100},"wireSpeed":{"min":0,"max":28,"unit":"m/min"},"min":0,"max":1102,"unit":"inch/min"}}}}$$$$
```

OTC(TIG):

```
####00000097{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":350},"wireSpeed":{"min":0,"max":28,"unit":"m/min"}}}}$$$$
```

Panasonic:

```
####000000d5{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":30,"max":500},"voltage":{"min":0,"max":65535},"voltageCompensation":{"min":-9.9,"max":9.9},"jobNumber":{"min":0,"max":255}}}}$$$$
```

Panasonic(RS485):

```
####00000085{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":30,"max":350},"voltage":{"min":5,"max":40}}}}$$$$
```

Riland:

```
####000000c3{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":500},"voltage":{"min":0,"max":50},"inductance":{"min":1,"max":50},"arcLength":{"min":-5,"max":5}}}}$$$$
```

Weco:

```
####000000e1{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":1,"max":99},"wireSpeed":[{"min":0,"max":6553.5,"unit":"m/min"}],"arcLength":{"min":0,"max":65535},"inductance":{"min":0,"max":65535}}}}$$$$
```

WeldTec(Ethernet/IP):

```
####000000b0{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"jobNumber":{"min":0,"max":99},"current":{"min":0,"max":1000},"voltageCompensation":{"min":-5,"max":5}}}}$$$$
```

WeldTec(ModbusTCP):

```
####000000ad{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"current":{"min":0,"max":500},"voltageCompensation":{"min":-5,"max":5},"voltage":{"min":0,"max":50}}}}$$$$
```

Hyperther:

```
####000000d9{"function":"DeviceConfig","parameter":{"type":"DeviceConfig","range":{"airPressure":{"min":58,"max":78},"current":{"min":20,"max":85},"cutterConfig":{"arcOnIO":"DO0","arcVoltageIO":"AI0","arcVoltageProportion":50}}}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "DeviceConfig",
  "parameter": {
    "type": "DeviceConfig",
    "ioConfig": {
      "AO0": {
        "type": 3,
        "curvePoints": [
          {
            "key": 0,
            "value": 0
          },
          {
            "key": 1,
            "value": 100
          },
          {
            "key": 10,
            "value": 500
          }
        ]
      }
    }
  }
},
```

```
"AO1": {
  "type": 3,
  "curvePoints": [
    {
      "key": 0,
      "value": -10
    },
    {
      "key": 2,
      "value": -5
    },
    {
      "key": 10,
      "value": 10
    }
  ]
},
"functionMap": {
  "outPut": {
    "arcOn": {
      "key": "CO1",
      "value": 0
    },
    "wirePush": {
      "key": "CO4",
      "value": 0
    },
    "wirePull": {
      "key": "CO2",
      "value": 0
    },
    "gasDetection": {
      "key": "",
      "value": 0
    },
    "enableTouchSense": {
      "key": "",
      "value": 0
    }
  },
  "inPut": {
    "welderReady": {
      "key": "",
      "value": 0
    },
  },
}
```

```
        "arcDetection": {  
            "key": "",  
            "value": 0  
        },  
        "touchSensed": {  
            "key": "",  
            "value": 0  
        },  
        "robotEmergencyStop": {  
            "key": "",  
            "value": 0  
        }  
    }  
}  
}
```

表 4-1-1 焊机配置-消息字段说明

参数		名称	数据类型	内容
function		功能类型	string	DeviceConfig
type		类型	string	DeviceConfig
range	current	电流范围	int	min 电流范围最小值
			int	max 电流范围最大值
	voltage	电压范围	int	min 电压范围最小值
			int	max 电压范围最大值
	voltageCompensation	电压补偿范围	int	min 电压补偿最小值
			int	max 电压补偿最大值
	arcLength	弧长校正	int	min 弧长校正最小值
			int	max 弧长校正最大值
	jobNumber	焊机 Job 号范围	int	min Job 号最小值
			int	max Job 号最大值
	powerCorrection	功率校正	int	min 功率校正最小值
			int	max 功率校正最大值
	interpulse	脉冲间隔	int	min 脉冲间隔最小值

			int	max 脉冲间隔最大值
	programNumber	程序号	int	min 程序号最小值
			int	max 程序号最大值
	unifyVoltage	一元化电压(OTC 焊机使用)	int	min 一元化电压最小值
			int	max 一元化电压最大值
	airPressure	切割气压(海宝等离子切割机使用)	int	min 切割气压最小值
			int	max 切割气压最大值
	laserDuty	激光占空比, 单位 % (IPG 激光焊机使用)	int	min 激光占空比最小值
			int	max 激光占空比最大值
	laserFrequency	激光频率, 单位 Hz (IPG 激光焊机使用)	int	min 激光频率最小值
			int	max 激光频率最大值
	swingFrequency	摆动频率, 单位 Hz (IPG 激光焊机使用)	int	min 摆动频率最小值
			int	max 摆动频率最大值
	swingLength	摆动宽度, 单位 mm (IPG 激光焊机使用)	int	min 摆动宽度最小值
			int	max 摆动宽度最大值
	wireSpeed	送丝速度, 类型: array	int	min 送丝速度最小值
			int	max 送丝速度最大值

			string	unit : m/min、inch/min、cm/min, 根据焊机支持情况选择,参考 发送消息
	cutterConfig (海宝等离子切割使用)	arcOnIO	string	起弧的 IO 位名称, 默认: D00
		arcVoltageIO	string	起弧电压输出, 默认: AI0
		arcVoltageProportion	int	弧压比例, 默认: 50

参数		名称	数据类型	内容
ioConfig	AO0	AO0 配置	object	模拟输出配置
	type	类型	number	曲线选项: 0:电流 (A), 1:送丝速度 (inch/min), 2:送丝速度 (m/min), 3:功率 (%)
	curvePoints	曲线点	array	[{ "key": 0, "value": 0 }, { "key": 1, "value": 100 }, { "key": 2, "value": 200 }, { "key": 10, "value": 500 }]
	AO1	AO1 配置	object	模拟输出配置
	type	类型	number	曲线选项: 0:电压 (V) 1:点压补偿 (V) 2:点压补偿 (%) 3:弧长校正
	curvePoints	曲线点	array	[{ "key": 0, "value": -10 }, { "key": 10, "value": 10 }]
	functionMap	功能映射	object	IO 功能映射配置

	outPut	输出映射	object	输出功能映射
	arcOn	起弧	object	"key": "CO0" ~ "CO7", "DO0" ~ "DO7", "value": 0 (低), 1 (高)
	wirePush	送丝	object	"key": "CO0" ~ "CO7", "DO0" ~ "DO7", "value": 0 (低), 1 (高)
	wirePull	退丝	object	"key": "CO0" ~ "CO7", "DO0" ~ "DO7", "value": 0 (低), 1 (高)
	gasDetection	送气检测	object	"key": "CO0" ~ "CO7", "DO0" ~ "DO7", "value": 0 (低), 1 (高)
	enableTouchSense	启用触碰感应	object	"key": "CO0" ~ "CO7", "DO0" ~ "DO7", "value": 0 (低), 1 (高)
	inPut	输入映射	object	输入功能映射
	welderReady	焊机准备	object	"key": "CI0" ~ "CI7", "DI0" ~ "DI7", "value": 0 (低), 1 (高)
	arcDetection	起弧检测	object	"key": "CI0" ~ "CI7", "DI0" ~ "DI7", "value": 0 (低), 1 (高)
	touchSensed	触碰感应	object	"key": "CI0" ~ "CI7", "DI0" ~ "DI7", "value": 0 (低), 1 (高)
	robotEmergencyStop	机器人急停	object	"key": "CI0" ~ "CI7", "DI0" ~ "DI7", "value": 0 (低), 1 (高)
	arcVoltageProportion	电弧电压比	double	null

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2 任务中心

“任务中心主要负责组织、调度和执行实际的机器人任务动作。它支持多种指令组合，最终通过统一的 runTask 指令下发至机器人进行执行。

在“运动”、“逻辑库”、“焊接库”等各子模块中，所有操作最终都通过 function: runTask 来调用。

一个任务的结构：由多个子任务，组成的集合体，比如：运动指令（如走直线、轨迹运动）；焊接参数（如电流电压、焊接模式）；条件逻辑（如 if-else，或信号触发）。

具体样例参考[第六章](#)

4.2.1 运动

4.2.1.1 描述：直线

发送消息：

```
####0000019d{"function":"runTask","parameter":{"id":"174410091144000000","name":"zhixian1","actions":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"174480747702800001","type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":"174480747702800000","type":"moveL"}],"type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": "174410091144000000",
    "name": "zhixian1",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 0,
              "y": 0,
              "z": 1176.5,
              "rx": 0,
```

```

        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": "174480747702800001",
    "type": "point"
  }
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": "174480747702800000",
"type": "moveL"
}
],
"type": 0
}
}

```

表 4-2-1 直线动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围：

			0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
actions[0]	id	string	动作 (action) ID
	speed	double	运动速度 (单位 mm/s)
	acceleration	double	运动加速度 (单位 mm/s ²)
	radius	double	过渡半径
	type	string	动作类型, 固定值: moveL
	children	array	点位列表见下表
children[0]	type	string	动作类型, 固定值: point
	id	string	动作 (action) ID
	enableVariable	int	启用寻位结果, “1” 使用, “0” 禁用
	variableSymbol	string	寻位结果
	position	object	点位信息, 见下表
position	x	double	点位 x 值 (单位: mm)
	y	double	点位 y 值 (单位: mm)
	z	double	点位 z 值 (单位: mm)
	rx	double	姿态 rx (单位: mm)

	ry	double	姿态 ry (单位: mm)
	rz	double	姿态 rz (单位: mm)
	J1	double	一轴角度 (单位: °)
	J2	double	二轴角度 (单位: °)
	J3	double	三轴角度 (单位: °)
	J4	double	四轴角度 (单位: °)
	J5	double	五轴角度 (单位: °)
	J6	double	六轴角度 (单位: °)
	TCP	string	当前 TCP 坐标系
	UCS	string	当前 UCS 坐标系

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 [附录 1](#) (下同)

4.2.1.2 描述: 关节

发送消息:

```
####0000019b{"function":"runTask","parameter":{"id":"174480788905100000","name":"guanjie",
,"actions":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J
4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"1744
807896400000001","type":"point"}],"speed":10,"acceleration":100,"radius":10,"id":"17448078964
0000000","type":"moveJ"}],"type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": "174480788905100000",
    "name": "guanjie",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 0,
              "y": 0,
              "z": 1176.5,
              "rx": 0,
              "ry": 0,
              "rz": 0,
              "J1": 0,
              "J2": 0,
              "J3": 0,
              "J4": 0,
              "J5": 0,
              "J6": 0,
              "TCP": "TCP",
              "UCS": "Base"
            },
            "enableVariable": 0,
            "variableSymbol": "p1",
            "id": "1744807896400000001",
            "type": "point"
          }
        ],
        "speed": 10,
        "acceleration": 100,
        "radius": 10,
        "id": "1744807896400000000",
        "type": "moveJ"
      }
    ]
  }
}
```

```

    ],
    "type": 0
  }
}

```

表 4-2-2 关节动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
actions[0]	id	string	动作 (action) ID
	speed	double	运动速度 (单位 mm/s)
	acceleration	double	运动加速度 (单位 mm/s ²)
	radius	double	过渡半径
	type	string	动作类型, 固定值: moveJ
	children	array	点位列表见下表
children[0]	type	string	动作类型, 固定值: point

	id	string	动作 (action) ID
	enableVariable	int	启用寻位结果, “1” 使用, “0” 禁用
	variableSymbol	string	寻位结果
	position	object	点位信息, 见下表
position	x	double	点位 x 值 (单位: mm)
	y	double	点位 y 值 (单位: mm)
	z	double	点位 z 值 (单位: mm)
	rx	double	姿态 rx (单位: mm)
	ry	double	姿态 ry (单位: mm)
	rz	double	姿态 rz (单位: mm)
	J1	double	一轴角度 (单位: °)
	J2	double	二轴角度 (单位: °)
	J3	double	三轴角度 (单位: °)
	J4	double	四轴角度 (单位: °)
	J5	double	五轴角度 (单位: °)
	J6	double	六轴角度 (单位: °)
	TCP	string	当前 TCP 坐标系

	UCS	string	当前 UCS 坐标系
--	-----	--------	------------

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.1.3 描述：圆弧

发送消息：

```
####0000029a{"function":"runTask","parameter":{"id":"174487563830200000","name":"circle","actions":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"174487564740500001","type":"point"},{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"174487564740600000","type":"point"}],"speed":100,"acceleration":500,"radius":10,"moveType":1,"radian":1,"fixedPosture":1,"id":"174487564740500000","type":"moveC"}],"type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": "174487563830200000",
    "name": "circle",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 0,
              "y": 0,
              "z": 1176.5,
```

```
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": "174487564740500001",
    "type": "point"
},
{
    "position": {
        "x": 0,
        "y": 0,
        "z": 1176.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": "174487564740600000",
    "type": "point"
}
},
"speed": 100,
"acceleration": 500,
"radius": 10,
"movecType": 1,
"radian": 1,
```

```

        "fixedPosture": 1,
        "id": "174487564740500000",
        "type": "moveC"
    }
],
"type": 0
}
}

```

表 4-2-3 圆弧动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
actions	动作列表	array	见下表
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions[0]	id	string	动作 (action) ID
	speed	double	运动速度 (单位 mm/s)
	acceleration	double	运动加速度 (单位 mm/s ²)
	radius	double	圆弧与后一动作间的过渡半径
	fixedPosture	int	固定姿态, 0: 不启用

			1: 启用
	movecType	int	类型, 固定值: 0: 整圆, 1: 圆弧
	radian	double	保留字段, 默认值: 1
	type	string	固定值: moveC
	children	array	点位列表见下表
children[0]	type	string	动作类型, 固定值: point
	id	string	动作 Id
	position	object	点位信息, 见下表
children[1]	参数	object	同 children[0]
position	x	double	点位 x 值 (单位: mm)
	y	double	点位 y 值 (单位: mm)
	z	double	点位 z 值 (单位: mm)
	rx	double	姿态 rx (单位: mm)
	ry	double	姿态 ry (单位: mm)
	rz	double	姿态 rz (单位: mm)
	J1	double	一轴角度 (单位: °)
	J2	double	二轴角度 (单位: °)

	J3	double	三轴角度（单位：°）
	J4	double	四轴角度（单位：°）
	J5	double	五轴角度（单位：°）
	J6	double	六轴角度（单位：°）
	TCP	string	当前 TCP 坐标系
	UCS	string	当前 UCS 坐标系

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.1.4 描述：摆动

发送消息：

```
####000006fa{"function":"runTask","parameter":{"id":"175159765170700000","name":"testwe","actions":[{"remarks":"","parentType":null,"children":[],"craftMode":2,"craftJob":null,"shape":0,"amplitude":2,"frequency":2,"dwellEnable":0,"dwellLeft":500,"dwellRight":500,"weaveFrameType":0,"elevation":0,"azimuth":0,"centralUplift":0,"suspendType":1,"weaveOffset":0,"taskId":"175159765170700000","id":"17515976517600000","order":1,"parentId":null,"device":"Megmeet_500_3","type":"weaveOn","isFirst":1,"nextSiblingId":"175159775808400000"},{"remarks":"","parentType":null,"children":[{"remarks":"","parentType":"moveL","children":[],"position":{"x":-1049.239,"y":-523.024,"z":592.259,"rx":170.051,"ry":-39.378,"rz":158.158,"J1":203.336,"J2":-23.931,"J3":63.595,"J4":2.717,"J5":96.194,"J6":214.507,"TCP":"TCP_1","UCS":"Base"},"jobNumber":1,"enableVariable":0,"variableSymbol":"p1","weldTemplateId":"","datumPoint":"","taskId":"175159765170700000"}
```

```
"id":"175159775808400001","order":1,"parentId":"175159775808400000","device":"Hansrobot_Elfin_0","type":"point","isFirst":"1","nextSiblingId":null}],{"speed":100,"acceleration":500,"radius":10,"taskId":"175159765170700000","id":"175159775808400000","order":2,"parentId":null,"device":"Hansrobot_Elfin_0","type":"moveL","isFirst":"0","nextSiblingId":"175159765917700000"},{"remarks":"","parentType":null,"children":[],"weaveOnId":"175159765917600000","taskId":"175159765170700000","id":"175159765917700000","order":1,"parentId":null,"device":"Hansrobot_Elfin_0","type":"weaveOff","isFirst":"0","nextSiblingId":null}],{"type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": "175159765170700000",
    "name": "testwe",
    "actions": [
      {
        "remarks": "",
        "parentType": null,
        "children": [],
        "craftMode": 2,
        "craftJob": null,
        "shape": 0,
        "amplitude": 2,
        "frequency": 2,
        "dwellEnable": 0,
        "dwellLeft": 500,
        "dwellRight": 500,
        "weaveFrameType": 0,
        "elevation": 0,
        "azimuth": 0,
        "centralUplift": 0,
        "suspendType": 1,
        "weaveOffId": "175159765917700000",
        "taskId": "175159765170700000",
        "id": "175159765917600000",
        "order": 1,
        "parentId": null,
        "device": "Megmeet_500_3",
        "type": "weaveOn",
        "isFirst": "1",
        "nextSiblingId": "175159775808400000"
      }
    ]
  }
}
```

```

{
  "remarks": "",
  "parentType": null,
  "children": [
    {
      "remarks": "",
      "parentType": "moveL",
      "children": [],
      "position": {
        "x": -1049.239,
        "y": -523.024,
        "z": 592.259,
        "rx": 170.051,
        "ry": -39.378,
        "rz": 158.158,
        "J1": 203.336,
        "J2": -23.931,
        "J3": 63.595,
        "J4": 2.717,
        "J5": 96.194,
        "J6": 214.507,
        "TCP": "TCP_1",
        "UCS": "Base"
      },
      "jobNumber": 1,
      "enableVariable": 0,
      "variableSymbol": "p1",
      "weldTemplateId": "",
      "datumPoint": "",
      "taskId": "175159765170700000",
      "id": "175159775808400001",
      "order": 1,
      "parentId": "175159775808400000",
      "device": "Hansrobot_Elfin_0",
      "type": "point",
      "isFirst": "1",
      "nextSiblingId": null
    }
  ],
  "speed": 100,
  "acceleration": 500,
  "radius": 10,
  "taskId": "175159765170700000",
  "id": "175159775808400000",
  "order": 2,

```



```

        "parentId": null,
        "device": "Hansrobot_Elfin_0",
        "type": "moveL",
        "isFirst": "0",
        "nextSiblingId": "175159765917700000"
    },
    {
        "remarks": "",
        "parentType": null,
        "children": [],
        "weaveOnId": "175159765917600000",
        "taskId": "175159765170700000",
        "id": "175159765917700000",
        "order": 1,
        "parentId": null,
        "device": "Hansrobot_Elfin_0",
        "type": "weaveOff",
        "isFirst": "0",
        "nextSiblingId": null
    }
],
"type": 0
}
}

```

表 4-2-4 摆动动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表

actions[0]	id	string	动作（action）ID
	type	string	动作类型，固定值：weaveOn
	shape	int	0 正弦摆动 2 三角摆动
	amplitude	double	摆动幅度
	frequency	double	频率
	dwelEnable	int	是否两端停留，固定值：0：否 1：是
	suspendType	int	停止类型，固定值：0：停留，1：前进
	dwelLeft	double	左端停留，停留时间，单位 ms，端侧停留有效，推荐 100ms 以上
	dwelRight	double	右端停留，停留时间，单位 ms，端侧停留有效，推荐 100ms 以上
	weaveFrameType	int	摆动平面，固定值：0：工具和焊道 1：工具
	centralUplift	int	中央隆起高度，单位 mm，三角摆动的时候，该值不能为 0
	elevation	int	仰角
	azimuth	int	方向角
	speed	double	前进速度
actions[1]	子动作列表	array	参考直线的 action 说明
actions[2]	id	string	动作（action）ID

	type	string	动作类型，固定值：weaveOff
--	------	--------	-------------------

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.1.5 描述：偏移

发送消息：

```
####00000658{"function":"runTask","parameter":{"id":174853810976200000,"name":"pi
anyi","actions":[{"offsetType":3,"offset":"o5","x":1,"y":2,"z":3,"rx":0,"ry":0,"rz":0,"id":174853
813146400000,"type":"offsetOn","children":[{"id":174853962644700000,"type":"teachOff
set","children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":
0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":17
4853962644700000,"type":"point"},{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1
":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variabl
eSymbol":"p1","id":174853962644700000,"type":"point"},{"position":{"x":0,"y":0,"z":1176.5,
"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"ena
bleVariable":0,"variableSymbol":"p1","id":174853962644700000,"type":"point"}]},{"id":174
853962644700000,"type":"teachOffset","children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,
"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVari
able":0,"variableSymbol":"p1","id":174853962644700000,"type":"point"},{"position":{"x":0,
"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UC
S":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174853962644700000,"type":"po
int"},{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"
J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174853962
644700000,"type":"point"}]},{id":174853813423000000,"type":"offsetOff"},"type":1}}$$
$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
```

```
"parameter": {
  "id": 174853810976200000,
  "name": "pianyi",
  "actions": [
    {
      "offsetType": 3,
      "offset": "o5",
      "x": 1,
      "y": 2,
      "z": 3,
      "rx": 0,
      "ry": 0,
      "rz": 0,
      "id": 174853813146400000,
      "type": "offsetOn",
      "children": [
        {
          "id": 174853962644700000,
          "type": "teachOffset",
          "children": [
            {
              "position": {
                "x": 0,
                "y": 0,
                "z": 1176.5,
                "rx": 0,
                "ry": 0,
                "rz": 0,
                "J1": 0,
                "J2": 0,
                "J3": 0,
                "J4": 0,
                "J5": 0,
                "J6": 0,
                "TCP": "TCP",
                "UCS": "Base"
              },
              "enableVariable": 0,
              "variableSymbol": "p1",
              "id": 174853962644700000,
              "type": "point"
            },
            {
              "position": {
                "x": 0,
```

```

        "y": 0,
        "z": 1176.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174853962644700000,
    "type": "point"
},
{
    "position": {
        "x": 0,
        "y": 0,
        "z": 1176.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174853962644700000,
    "type": "point"
}
]
},
{
    "id": 174853962644700000,

```

```
"type": "teachOffset",
"children": [
  {
    "position": {
      "x": 0,
      "y": 0,
      "z": 1176.5,
      "rx": 0,
      "ry": 0,
      "rz": 0,
      "J1": 0,
      "J2": 0,
      "J3": 0,
      "J4": 0,
      "J5": 0,
      "J6": 0,
      "TCP": "TCP",
      "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174853962644700000,
    "type": "point"
  },
  {
    "position": {
      "x": 0,
      "y": 0,
      "z": 1176.5,
      "rx": 0,
      "ry": 0,
      "rz": 0,
      "J1": 0,
      "J2": 0,
      "J3": 0,
      "J4": 0,
      "J5": 0,
      "J6": 0,
      "TCP": "TCP",
      "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174853962644700000,
    "type": "point"
  }
]
```

```

    },
    {
      "position": {
        "x": 0,
        "y": 0,
        "z": 1176.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174853962644700000,
      "type": "point"
    }
  ]
},
{
  "id": 174853813423000000,
  "type": "offsetOff"
}
],
"type": 1
}
}

```

表 4-2-5 偏移动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表

id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围：0：模拟任务，不起弧；1：真实任务，起弧；
actions	动作列表	array	见下表
actions[0]	id	string	动作（action）ID
	type	string	动作类型，固定值：offsetOn
	offsetType	int	偏移类型：0:手动偏移，1:接触寻位偏移，2:激光寻位偏移，3:示教偏移
	offset	string	激光寻位偏移值
	x	int	手动偏移：x 轴（单位 mm）
	y	int	手动偏移：y 轴（单位 mm）
	z	int	手动偏移：z 轴（单位 mm）
	children[0]	position	见 4.2.1.1（position 字段）
		enableVariable	是否使用寻位结果
		variableSymbol	寻位结果
		id	动作 id
		type	point
actions[1]	type	string	动作类型，固定值：offsetOff

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.1.6 描述：相贯线

发送消息：

```
####000003c8{"function":"runTask","parameter":{"id":174796997168900000,"name":"test","actions":[{"children":[{"position":{"x":428.455,"y":10.97,"z":443.373,"rx":-179.979,"ry":-0.825,"rz":-180,"J1":1.467,"J2":-1.636,"J3":88.362,"J4":0,"J5":90.827,"J6":1.466,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174796998216900000,"type":"point"},{"position":{"x":428.455,"y":10.97,"z":443.373,"rx":-179.979,"ry":-0.825,"rz":-180,"J1":1.467,"J2":-1.636,"J3":88.362,"J4":0,"J5":90.827,"J6":1.466,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174796998217000000,"type":"point"},{"position":{"x":428.455,"y":10.97,"z":443.373,"rx":-179.979,"ry":-0.825,"rz":-180,"J1":1.467,"J2":-1.636,"J3":88.362,"J4":0,"J5":90.827,"J6":1.466,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174796998217000000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174796998216900000,"type":"moveIntersecting"}], "type":1}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": 174796997168900000,
    "name": "test",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 428.455,
              "y": 10.97,
              "z": 443.373,
```

```
    "rx": -179.979,  
    "ry": -0.825,  
    "rz": -180,  
    "J1": 1.467,  
    "J2": -1.636,  
    "J3": 88.362,  
    "J4": 0,  
    "J5": 90.827,  
    "J6": 1.466,  
    "TCP": "TCP",  
    "UCS": "Base"  
  },  
  "enableVariable": 0,  
  "variableSymbol": "p1",  
  "id": 174796998216900000,  
  "type": "point"  
},  
{  
  "position": {  
    "x": 428.455,  
    "y": 10.97,  
    "z": 443.373,  
    "rx": -179.979,  
    "ry": -0.825,  
    "rz": -180,  
    "J1": 1.467,  
    "J2": -1.636,  
    "J3": 88.362,  
    "J4": 0,  
    "J5": 90.827,  
    "J6": 1.466,  
    "TCP": "TCP",  
    "UCS": "Base"  
  },  
  "enableVariable": 0,  
  "variableSymbol": "p1",  
  "id": 174796998217000000,  
  "type": "point"  
},  
{  
  "position": {  
    "x": 428.455,  
    "y": 10.97,  
    "z": 443.373,  
    "rx": -179.979,
```

```

        "ry": -0.825,
        "rz": -180,
        "J1": 1.467,
        "J2": -1.636,
        "J3": 88.362,
        "J4": 0,
        "J5": 90.827,
        "J6": 1.466,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174796998217000000,
    "type": "point"
  }
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": 174796998216900000,
"type": "moveIntersecting"
}
],
"type": 1
}
}

```

表 4-2-6 相贯线动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称

type	任务类型	int	取值范围：0：模拟任务，不起弧；1：真实任务，起弧；
actions	动作列表	array	见下表
actions[0]	id	string	动作（action）ID
	type	string	动作类型，固定值：moveIntersecting
	speed	double	工具速度（单位：mm/s）
	acceleration	double	工具加速度（单位：mm/s ² ）
	radius	int	交融半径（单位：mm）
	children[0]	position	见 4.2.1.1（position 字段）
		enableVariable	是否使用寻位结果
		variableSymbol	寻位结果
		id	动作 id
		type	point
actions[1]	子动作列表	array	参考直线的 action 说明
actions[2]	id	string	动作（action）ID
	type	string	动作类型，固定值：weaveOff

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.1.7 描述：接触寻位

发送消息:

一维平移:

```
####000004af{"function":"runTask","parameter":{"id":"177083067401400000","name":"f2","actions":[{"children":[{"children":[{"position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"enableVariable":0,"variableSymbol":"","id":"177226427149300002","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177226427152300001","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177226427152300002","type":"point"}],"speed":10,"acceleration":2500,"distance":200,"id":"177226427149300001","type":"positioning"}],"positioningType":0,"id":"177226427149300000","type":"touchSense"},"type":1}}$$$$
```

二维平移:

```
####000007e3{"function":"runTask","parameter":{"id":"177083067401400000","name":"f2","actions":[{"positioningType":1,"id":"177226427149300000","type":"touchSense","children":[{"speed":10,"acceleration":2500,"distance":200,"id":"177226721533100001","type":"positioning","children":[{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177226721540800002","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177226721540800003","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177226721540800004","type":"point"}]},{"disabled":0,"speed":10,"acceleration":2500,"distance":200,"id":"177226721540800005","type":"positioning","children":[{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177226721540800009","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"B
```

```
ase"},"id":"177226721540800010","type":"point"),{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177226721540800011","type":"point"}]]]],"type":1}}$$$$
```

二维平移+旋转

```
####00000e3e{"function":"runTask","parameter":{"id":"177083067401400000","name":"f2","actions":[{"positioningType":2,"id":"177226427149300000","type":"touchSense","children":[{"speed":10,"acceleration":2500,"distance":200,"id":"177227032401100001","type":"positioning","children":[{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400002","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400003","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400004","type":"point"}]},{"disabled":0,"speed":10,"acceleration":2500,"distance":200,"id":"177227032405400005","type":"positioning","children":[{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400009","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400010","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400011","type":"point"}]},{"speed":10,"acceleration":2500,"distance":200,"id":"177227032405400012","type":"positioning","children":[{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400016","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400017","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405400018","type":"point"}]},{"disabled":0,"speed":10,"acceleration":2500,"distance":200,"id":"177227032405400019","type":"positioning","children":[{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405500002","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405500003","type":"point"},{"enableVariable":0,"variableSymbol":"","position":{"x":-18.184,"y":-8.397,"z":1765.588,"rx":147.999,"ry":0,"rz":-67.501,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP_1","UCS":"Base"},"id":"177227032405500004","type":"point"}]]]],"type":1}}$$$$
```

消息体 JSON 格式化:

一维平移

```
{
  "function": "runTask",
  "parameter": {
    "id": "177083067401400000",
    "name": "f2",
    "actions": [
      {
        "children": [
          {
            "children": [
              {
                "position": {
                  "x": -18.184,
                  "y": -8.397,
                  "z": 1765.588,
                  "rx": 147.999,
                  "ry": 0,
                  "rz": -67.501,
                  "J1": 0,
                  "J2": 0,
                  "J3": 0,
                  "J4": 0,
                  "J5": 0,
                  "J6": 0,
                  "TCP": "TCP_1",
                  "UCS": "Base"
                },
                "enableVariable": 0,
                "variableSymbol": "",
                "id": "177226427149300002",
                "type": "point"
              },
              {
                "enableVariable": 0,
                "variableSymbol": "",
                "position": {
                  "x": -18.184,
                  "y": -8.397,
```

```

        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177226427152300001",
    "type": "point"
},
{
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177226427152300002",
    "type": "point"
}
],
"speed": 10,
"acceleration": 2500,
"distance": 200,
"id": "177226427149300001",
"type": "positioning"
}

```



```

    ],
    "positioningType": 0,
    "id": "177226427149300000",
    "type": "touchSense"
  }
],
  "type": 1
}
}

```

二维平移

```

{
  "function": "runTask",
  "parameter": {
    "id": "177083067401400000",
    "name": "f2",
    "actions": [
      {
        "positioningType": 1,
        "id": "177226427149300000",
        "type": "touchSense",
        "children": [
          {
            "speed": 10,
            "acceleration": 2500,
            "distance": 200,
            "id": "177226721533100001",
            "type": "positioning",
            "children": [
              {
                "enableVariable": 0,
                "variableSymbol": "",
                "position": {
                  "x": -18.184,
                  "y": -8.397,
                  "z": 1765.588,
                  "rx": 147.999,
                  "ry": 0,
                  "rz": -67.501,
                  "J1": 0,
                  "J2": 0,
                  "J3": 0,
                  "J4": 0,
                  "J5": 0,
                  "J6": 0,

```

```

        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177226721540800002",
    "type": "point"
},
{
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177226721540800003",
    "type": "point"
},
{
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",

```

```

        "UCS": "Base"
      },
      "id": "177226721540800004",
      "type": "point"
    }
  ]
},
{
  "disabled": 0,
  "speed": 10,
  "acceleration": 2500,
  "distance": 200,
  "id": "177226721540800005",
  "type": "positioning",
  "children": [
    {
      "enableVariable": 0,
      "variableSymbol": "",
      "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
      },
      "id": "177226721540800009",
      "type": "point"
    },
    {
      "enableVariable": 0,
      "variableSymbol": "",
      "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,

```

```

    "ry": 0,
    "rz": -67.501,
    "J1": 0,
    "J2": 0,
    "J3": 0,
    "J4": 0,
    "J5": 0,
    "J6": 0,
    "TCP": "TCP_1",
    "UCS": "Base"
  },
  "id": "177226721540800010",
  "type": "point"
},
{
  "enableVariable": 0,
  "variableSymbol": "",
  "position": {
    "x": -18.184,
    "y": -8.397,
    "z": 1765.588,
    "rx": 147.999,
    "ry": 0,
    "rz": -67.501,
    "J1": 0,
    "J2": 0,
    "J3": 0,
    "J4": 0,
    "J5": 0,
    "J6": 0,
    "TCP": "TCP_1",
    "UCS": "Base"
  },
  "id": "177226721540800011",
  "type": "point"
}
]
}
],
"type": 1
}
}
二维平移+旋转
```

```

{
  "function": "runTask",
  "parameter": {
    "id": "177083067401400000",
    "name": "f2",
    "actions": [
      {
        "positioningType": 2,
        "id": "177226427149300000",
        "type": "touchSense",
        "children": [
          {
            "speed": 10,
            "acceleration": 2500,
            "distance": 200,
            "id": "177227032401100001",
            "type": "positioning",
            "children": [
              {
                "enableVariable": 0,
                "variableSymbol": "",
                "position": {
                  "x": -18.184,
                  "y": -8.397,
                  "z": 1765.588,
                  "rx": 147.999,
                  "ry": 0,
                  "rz": -67.501,
                  "J1": 0,
                  "J2": 0,
                  "J3": 0,
                  "J4": 0,
                  "J5": 0,
                  "J6": 0,
                  "TCP": "TCP_1",
                  "UCS": "Base"
                },
                "id": "177227032405400002",
                "type": "point"
              },
              {
                "enableVariable": 0,
                "variableSymbol": "",
                "position": {
                  "x": -18.184,

```

```

        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177227032405400003",
    "type": "point"
},
{
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177227032405400004",
    "type": "point"
}
]
},
{
    "disabled": 0,
    "speed": 10,
    "acceleration": 2500,

```

```

"distance": 200,
"id": "177227032405400005",
"type": "positioning",
"children": [
  {
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
      "x": -18.184,
      "y": -8.397,
      "z": 1765.588,
      "rx": 147.999,
      "ry": 0,
      "rz": -67.501,
      "J1": 0,
      "J2": 0,
      "J3": 0,
      "J4": 0,
      "J5": 0,
      "J6": 0,
      "TCP": "TCP_1",
      "UCS": "Base"
    },
    "id": "177227032405400009",
    "type": "point"
  },
  {
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
      "x": -18.184,
      "y": -8.397,
      "z": 1765.588,
      "rx": 147.999,
      "ry": 0,
      "rz": -67.501,
      "J1": 0,
      "J2": 0,
      "J3": 0,
      "J4": 0,
      "J5": 0,
      "J6": 0,
      "TCP": "TCP_1",
      "UCS": "Base"
    },
  },

```

```

        "id": "177227032405400010",
        "type": "point"
    },
    {
        "enableVariable": 0,
        "variableSymbol": "",
        "position": {
            "x": -18.184,
            "y": -8.397,
            "z": 1765.588,
            "rx": 147.999,
            "ry": 0,
            "rz": -67.501,
            "J1": 0,
            "J2": 0,
            "J3": 0,
            "J4": 0,
            "J5": 0,
            "J6": 0,
            "TCP": "TCP_1",
            "UCS": "Base"
        },
        "id": "177227032405400011",
        "type": "point"
    }
]
},
{
    "speed": 10,
    "acceleration": 2500,
    "distance": 200,
    "id": "177227032405400012",
    "type": "positioning",
    "children": [
        {
            "enableVariable": 0,
            "variableSymbol": "",
            "position": {
                "x": -18.184,
                "y": -8.397,
                "z": 1765.588,
                "rx": 147.999,
                "ry": 0,
                "rz": -67.501,
                "J1": 0,

```



```

        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177227032405400016",
    "type": "point"
},
{
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177227032405400017",
    "type": "point"
},
{
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
        "x": -18.184,
        "y": -8.397,
        "z": 1765.588,
        "rx": 147.999,
        "ry": 0,
        "rz": -67.501,
        "J1": 0,
        "J2": 0,

```

```

        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP_1",
        "UCS": "Base"
    },
    "id": "177227032405400018",
    "type": "point"
}
]
},
{
    "disabled": 0,
    "speed": 10,
    "acceleration": 2500,
    "distance": 200,
    "id": "177227032405400019",
    "type": "positioning",
    "children": [
        {
            "enableVariable": 0,
            "variableSymbol": "",
            "position": {
                "x": -18.184,
                "y": -8.397,
                "z": 1765.588,
                "rx": 147.999,
                "ry": 0,
                "rz": -67.501,
                "J1": 0,
                "J2": 0,
                "J3": 0,
                "J4": 0,
                "J5": 0,
                "J6": 0,
                "TCP": "TCP_1",
                "UCS": "Base"
            },
            "id": "177227032405500002",
            "type": "point"
        },
        {
            "enableVariable": 0,
            "variableSymbol": "",

```

```

    "position": {
      "x": -18.184,
      "y": -8.397,
      "z": 1765.588,
      "rx": 147.999,
      "ry": 0,
      "rz": -67.501,
      "J1": 0,
      "J2": 0,
      "J3": 0,
      "J4": 0,
      "J5": 0,
      "J6": 0,
      "TCP": "TCP_1",
      "UCS": "Base"
    },
    "id": "177227032405500003",
    "type": "point"
  },
  {
    "enableVariable": 0,
    "variableSymbol": "",
    "position": {
      "x": -18.184,
      "y": -8.397,
      "z": 1765.588,
      "rx": 147.999,
      "ry": 0,
      "rz": -67.501,
      "J1": 0,
      "J2": 0,
      "J3": 0,
      "J4": 0,
      "J5": 0,
      "J6": 0,
      "TCP": "TCP_1",
      "UCS": "Base"
    },
    "id": "177227032405500004",
    "type": "point"
  }
]
}

```

```

    ],
    "type": 1
  }
}

```

表 4-2-7 接触寻位动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
actions[0]	id	string	动作 (action) ID
	type	string	动作类型, 固定值: touchSense
	positioningType	int	寻位类型 (0:一维平移, 1:二维平移, 2:二维平移旋转)
	children[0].children	speed	工具速度
		acceleration	加速度
		distance	寻位距离
		id	动作 id

		type	固定值: positioning 一维平移: 3 个点位, 顺序为: 起点、基准点、寻位成功返回点 二维平移: 寻位两个面, 6 个点位, 每个面 3 个点位, 顺序为: 起点、基准点、寻位成功返回点 二维平移+旋转: 寻位两个面, 每个面寻位两次, 12 个点位, 每次 3 个点位, 顺序为: 起点、基准点、寻位成功返回点
		position	见 4.2.1.1 (position 字段)
		enableVariable	是否使用寻位结果, 固定值: 0, 暂不使用
		variableSymbol	寻位结果, 固定值: "", 暂不使用
		id	动作 id
		type	point

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.2 逻辑库

4.2.2.1 描述：If 判断（不支持嵌套）

发送消息：

```
####00000606{"function":"runTask","parameter":{"id":"174487743628300000","name":"if",
,"actions":[{"children":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":
0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variable
Symbol":"p1","id":"174487798868900001","type":"point"}],"speed":100,"acceleration":500,
"radius":10,"id":"174487798868900000","type":"moveL"}],"logic":{"type":"readlo","express
ion":[{"operand":"Cl1","operator":"=","value":0}],"ifEndId":"174487745239300001","id":"1
74487745239300000","type":"ifStart"},{"logic":{"type":"readlo","expression":[{"operand":"
Cl2","operator":"=","value":1}],"ifStartId":"174487745239300000","id":"17448775364000
0000","type":"elseif","children":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"r
z":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,
"variableSymbol":"p1","id":"174487799185400001","type":"point"}],"speed":100,"accelera
tion":500,"radius":10,"id":"174487799185400000","type":"moveL"}],"ifStartId":"1744877
45239300000","id":"174487755280800000","type":"else","children":[{"children":[{"positio
n":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"
TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"174487799407500001
","type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":"174487799407500000",
"type":"moveL"}]},"ifStartId":"174487745239300000","id":"174487745239300001","type
":"ifEnd"},"type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": "174487743628300000",
    "name": "if",
    "actions": [
      {
        "children": [
          {
            "children": [
              {
                "position": {
                  "x": 0,
                  "y": 0,
                  "z": 1176.5,
                  "rx": 0,
                  "ry": 0,
```

```

        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": "174487798868900001",
    "type": "point"
}
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": "174487798868900000",
"type": "moveL"
}
],
"logic": {
    "type": "readlo",
    "expression": [
        {
            "operand": "Cl1",
            "operator": "=",
            "value": 0
        }
    ]
},
    "ifEndId": "174487745239300001"
    "id": "174487745239300000",
    "type": "ifStart"
},
{
    "logic": {
        "type": "readlo",
        "expression": [
            {
                "operand": "Cl2",
                "operator": "=",
                "value": 1
            }
        ]
    }
}

```

```

    }
  ]
},
"ifStartId": "174487745239300000"
"id": "174487753640000000",
"type": "elseif",
"children": [
  {
    "children": [
      {
        "position": {
          "x": 0,
          "y": 0,
          "z": 1176.5,
          "rx": 0,
          "ry": 0,
          "rz": 0,
          "J1": 0,
          "J2": 0,
          "J3": 0,
          "J4": 0,
          "J5": 0,
          "J6": 0,
          "TCP": "TCP",
          "UCS": "Base"
        },
        "enableVariable": 0,
        "variableSymbol": "p1",
        "id": "174487799185400001",
        "type": "point"
      }
    ],
    "speed": 100,
    "acceleration": 500,
    "radius": 10,
    "id": "174487799185400000",
    "type": "moveL"
  }
]
},
{
  "ifStartId": "174487745239300000"
  "id": "174487755280800000",
  "type": "else",
  "children": [

```



```

        {
            "children": [
                {
                    "position": {
                        "x": 0,
                        "y": 0,
                        "z": 1176.5,
                        "rx": 0,
                        "ry": 0,
                        "rz": 0,
                        "J1": 0,
                        "J2": 0,
                        "J3": 0,
                        "J4": 0,
                        "J5": 0,
                        "J6": 0,
                        "TCP": "TCP",
                        "UCS": "Base"
                    },
                    "enableVariable": 0,
                    "variableSymbol": "p1",
                    "id": "174487799407500001",
                    "type": "point"
                }
            ],
            "speed": 100,
            "acceleration": 500,
            "radius": 10,
            "id": "174487799407500000",
            "type": "moveL"
        }
    ],
    {
        "ifStartId": "174487745239300000"
        "id": "174487745239300001",
        "type": "ifEnd"
    }
],
"type": 0
}

```

表 4-2-5 if 动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	动作类型	string	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表

actions[0]	id		string	动作 (action) ID
	type	动作类型	string	固定值: ifStart
	logic expression	type	string	固定值: readlo
		operand	string	取值范围: CI0,CI1,CI2,CI3... DI0,DI1,DI2,DI3...
		operator	string	固定值: =
		value	int	取值范围: 0: 低 1: 高
	children	子动作列表	array	见其他动作 action

actions[1]	id		string	动作（action）ID
	type	动作 action	string	固定值：elseif
	logic	type		固定值：readlo
		expression	operand	取值范围： CI0,CI1,CI2,CI3... DI0,DI1,DI2,DI3...
			operator	固定值：=
		value	int	取值范围：0：低 1：高

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 [附录 1](#)

4.2.2.2 描述：循环（不支持嵌套）

发送消息：

```
####000002a3{"function":"runTask","parameter":{"id":"174487819575700000","name":"loop","actions":[{"children":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"174487867641700001","type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":"174487867641700000","type":"moveL"}],"logic":{"type":"times","expression":[{"operand":"CI0","operator":"=","value":0}],"times":1},"loopEndId":"174487865646700000","id":"174487865646600000","type":"loopStart"},{"loopStartId":"174487865646600000","id":"174487865646700000","type":"loopEnd"},"type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": "174487819575700000",
    "name": "loop",
    "actions": [
      {
        "children": [
          {
            "children": [
              {
                "position": {
                  "x": 0,
                  "y": 0,
                  "z": 1176.5,
                  "rx": 0,
                  "ry": 0,
                  "rz": 0,
                  "J1": 0,
                  "J2": 0,
                  "J3": 0,
                  "J4": 0,
                  "J5": 0,
                  "J6": 0,
                  "TCP": "TCP",
                  "UCS": "Base"
                },
                "enableVariable": 0,
                "variableSymbol": "p1",
                "id": "174487867641700001",
                "type": "point"
              }
            ],
            "speed": 100,
            "acceleration": 500,
            "radius": 10,
            "id": "174487867641700000",
            "type": "moveL"
          }
        ],
        "logic": {
          "type": "times",
          "expression": [
            {
```

```

        "operand": "C10",
        "operator": "=",
        "value": 0
      }
    ],
    "times": 1
  },
  "loopEndId": "174487865646700000",
  "id": "174487865646600000",
  "type": "loopStart"
},
{
  "loopStartId": "174487865646600000",
  "id": "174487865646700000",
  "type": "loopEnd"
}
],
"type": 0
}
}

```

表 4-2-6 loop 动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	动作类型	string	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions[0]	id	string	动作 (action) ID

	type	动作类型		string	固定值: loopStart
	logic	type		string	取值范围: times: 循环次数 readlo: 读取 IO
		expression	operand	string	取值范围: CI0,CI1,CI2,CI3... DI0,DI1,DI2,DI3...
			operator	string	固定值: =
			value	int	取值范围: 0: 低 1: 高
		times		int	循环次数
	children	子动作列表		array	见其他动作 action

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 [附录 1](#)

4.2.2.3 描述：等待

发送消息:

```
####00000237{"function":"runTask","parameter":{"id":"174487911512200000","name":"wait","actions":[{"logic":{"type":"delay","expression":[{"operand":"timeout","operator":"=","v
```

```

    "value":200}},"id":"174487912392700000","type":"wait"},{"children":[{"position":{"x":145.85
7,"y":0,"z":921.368,"rx":0,"ry":62.858,"rz":0,"J1":0,"J2":39.749,"J3":76.206,"J4":0,"J5":26.4
01,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"17448
7914149700001","type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":"174487
914149700000","type":"moveL"},"type":0}}$$$$

```

消息体 JSON 格式化:

```

{
  "function": "runTask",
  "parameter": {
    "id": "174487911512200000",
    "name": "wait",
    "actions": [
      {
        "logic": {
          "type": "delay",
          "expression": [
            {
              "operand": "timeout",
              "operator": "=",
              "value": 200
            }
          ]
        }
      }
    ],
    "id": "174487912392700000",
    "type": "wait"
  },
  {
    "children": [
      {
        "position": {
          "x": 145.857,
          "y": 0,
          "z": 921.368,
          "rx": 0,
          "ry": 62.858,
          "rz": 0,
          "J1": 0,
          "J2": 39.749,
          "J3": 76.206,
          "J4": 0,
          "J5": 26.401,
          "J6": 0,
          "TCP": "TCP",

```

```

        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": "174487914149700001",
      "type": "point"
    }
  ],
  "speed": 100,
  "acceleration": 500,
  "radius": 10
  "id": "174487914149700000",
  "type": "moveL"
}
],
"type": 0
}
}

```

表 4-2-7 wait 动作-消息字段说明

参数	名称		数据类型	内容
function	功能名称		string	固定值: runTask
parameter	参数		object	见下表
id	任务 ID		string	任务 ID
name	任务名称		string	任务名称
type	动作类型		string	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions[0]	id		string	动作 (action) ID
	type	动作类型	string	固定值: wait

	logic	type		string	取值范围： delay: 循环次数 readlo: IO 变化
		expression	operand	string	固定值: timeout
			operator	string	固定值: =
			value	int	取值范围： 0 以上
		times		int	循环次数

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.2.4 描述: 设置 IO 信号

发送消息:

```
####000001fb{"function":"runTask","parameter":{"id":"174487928376500000","name":"setIO","actions":[{"ioName":"COO","ioState":1,"id":"174487929418800000","type":"setio"},{"children":[{"position":{"x":145.857,"y":0,"z":921.368,"rx":0,"ry":62.858,"rz":0,"J1":0,"J2":39.749,"J3":76.206,"J4":0,"J5":26.401,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":"174487930331600001","type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":"174487930331600000","type":"moveL"}],"type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
```

```

"parameter": {
  "id": "174487928376500000",
  "name": "setIO",
  "actions": [
    {
      "ioName": "CO0",
      "ioState": 1,
      "id": "174487929418800000",
      "type": "setio"
    },
    {
      "children": [
        {
          "position": {
            "x": 145.857,
            "y": 0,
            "z": 921.368,
            "rx": 0,
            "ry": 62.858,
            "rz": 0,
            "J1": 0,
            "J2": 39.749,
            "J3": 76.206,
            "J4": 0,
            "J5": 26.401,
            "J6": 0,
            "TCP": "TCP",
            "UCS": "Base"
          },
          "enableVariable": 0,
          "variableSymbol": "p1",
          "id": "174487930331600001",
          "type": "point"
        }
      ],
      "speed": 100,
      "acceleration": 500,
      "radius": 10,
      "id": "174487930331600000",
      "type": "moveL"
    }
  ],
  "type": 0
}

```

表 4-2-8 设置 IO 信号-消息字段说明

参数	名称		数据类型	内容
function	功能名称		string	固定值: runTask
id	任务 ID		int	任务 ID
name	任务名称		string	任务名称
type	动作类型		string	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions[0]	id		string	动作 (action) ID
	type	动作类型	string	固定值: setio
	ioName		string	取值范围: CI0,CI1,CI2,CI3...DI0,DI1,DI2,DI3...
	ioState		int	固定值: 0: 低 1: 高

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3 焊接库

4.2.3.1 描述：起弧

发送消息：

```
####000004a1{"function":"runTask","parameter":{"id":"174487947880600000","name":"arcOn","actions":[{"arcOn":{"arcOnMode":2,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnTime":200,"preflowTime":100,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0},"arcOff":{"arcOffMode":3,"arcOffCurrent":100,"arcOffVoltage":30,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0},"arcSet":{"arcSetMode":1,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":20,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":"174487949033400000","type":"arcOn","arcOnClass":0},"type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": "174487947880600000",
    "name": "arcOn",
    "actions": [
      {
        "arcOn": {
          "arcOnMode": 2,
          "arcOnCurrent": 100,
          "arcOnAirPressure": 0,
          "arcOnVoltage": 0,
          "arcOnTime": 200,
          "preflowTime": 100,
          "arcOnPowerCorrection": 20,
          "arcOnWireSpeed": 0,
          "arcOnJobNumber": 1,
          "arcOnInductance": 0,
          "arcOnSlope": 0,
          "arcOnArcControl": 0
        },
        "arcOff": {
          "arcOffMode": 3,
```

```
"arcOffCurrent": 100,
"arcOffVoltage": 30,
"arcOffTime": 0,
"arcOffDistance": 0,
"enableArcOffDistance": true,
"postflowTime": 0,
"enableBurnback": 1,
"burnBackTime": 0,
"arcOffPowerCorrection": 20,
"arcOffWireSpeed": 0,
"arcOffJobNumber": 1,
"arcOffInductance": 0,
"arcOffSlope": 0,
"arcOffArcControl": 0
},
"arcSet": {
  "arcSetMode": 1,
  "arcSetCurrent": 100,
  "arcSetAirPressure": 0,
  "arcSetVoltage": 20,
  "arcSetPowerCorrection": 20,
  "arcSetWireSpeed": 0,
  "arcSetJobNumber": 1,
  "arcSetInductance": 0,
  "arcSetScanSpeed": 200,
  "arcSetScanLength": 2,
  "arcSetSpduty": 50,
  "arcSetSpfreq": 2000,
  "arcSetSwingLength": 0,
  "arcSetSwingFrequency": 0,
  "speed": 10,
  "enablePushWire": 0,
  "arcSetSlope": 0,
  "arcSetArcControl": 0
},
"weldProgram": {
  "isUseMemorySlot": 0,
  "programNumber": 1,
  "weldProcess": 0,
  "weldWireType": 0,
  "weldWireSize": 0,
  "weldGasType": 0
},
"id": "174487949033400000",
"type": "arcOn",
```

```

        "arcOnClass": 0
    }
],
    "type": 0
}
}

```

表 4-2-9 起弧动作-消息字段说明

参数	名称	数据类型	内容	
function	功能名称	string	固定值: runTask	
parameter	参数	object	见下表	
id	任务 ID	string	任务 ID	
name	任务名称	string	任务名称	
type	任务类型	string	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;	
actions	动作列表	array	见下表	
actions[0]	起弧	object	见下表	
actions[1]	id	动作 (action) ID	int	动作 (action) ID
	type	动作类型	string	固定值: arcOn
	arcOn	起弧工艺	object	见下表
	arcOnMode	起弧模式	int	见 5.3
	arcOnCurrent	起弧/切割电流	int	

	arcOnVoltage	起弧电压	double	
	arcOnTime	起弧时间	int	
	preflowTime	提前送气	int	
	arcOnPowerCorrection	起弧功率校正	int	
	arcOnWireSpeed	送丝速度	int	
	arcOnJobNumber	焊接模式	int	见 5.3
	arcOnAirPressure	切割气压	double	海宝使用
	arcOnArcControl	电弧控制	double	
	arcOnInductance	设置电感	int	
	arcOnSlope	设置斜率	double	
	arcOnSharpArc	设置尖锐弧	int	
	arcOff	收弧工艺	object	见下表
	arcOffMode	收弧模式	int	见 5.3
	arcOffDistance	衰减距离	double	
	enableArcOffDistance	是否启用衰减 距离	int	1:启用, 0:禁用
	arcOffArcControl	电弧控制	double	
	arcOffCurrent	收弧电流	int	

	arcOffVoltage	收弧电压	double	
	arcOffTime	收弧时间	int	
	postflowTime	延后送气	int	
	enableBurnback	是否启用退丝	bool	
	burnBackTime	退丝时间	int	
	arcOffPowerCorrection	收弧功率校正	int	
	arcSet	焊接工艺	object	见下表
	arcSetCurrent	焊接/切割电流	int	
	arcSetVoltage	焊接电压	double	
	arcSetWireSpeed	送丝速度	int	
	arcSetJobNumber	焊接模式	int	见 5.3
	speed	焊接速度	int	
	arcSetInductance	电感	int	
	arcSetSlope	坡度	int	
	arcSetAirPressure	切割气压	double	海宝使用
	arcSetScanSpeed	扫描速度	double	
	arcSetScanLength	扫描长度	double	

	arcSetSpduty	占空比	double	
	arcSetSpfreq	频率	double	
	arcSetSwingLength	摆动长度	double	
	arcSetSwingFrequency	摆动频率	double	
	speed	速度	double	
	enablePushWire	启用送丝	number	0: 禁用, 1:启用
	arcSetSlope	焊接斜率	double	
	arcSetArcControl	焊接弧长控制	double	
	arcOnClass	起弧类别	number	0: 立即起弧, 1:延迟起弧
	weldProgram	焊接程序	object	见 4.1.20 (仅米勒 ModBus TCP/IP 模式)

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.2 描述: 收弧

发送消息：

```
####00000090{"function":"runTask","parameter":{"id":"171023527750479002","name":"arc_off","type":0,"actions":[{"id":"171023551228370385","type":"arcOff"}]}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": "171023527750479002",
    "name": "arc_off",
    "type": 0,
    "actions": [
      {
        "id": "171023551228370385",
        "type": "arcOff"
      }
    ]
  }
}
```

表 4-2-10 收弧动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	string	取值范围： 0：模拟任务，不起弧； 1：真实任务，起弧；
actions	动作列表	array	见下表

actions[0]	收弧	object	见下表
id	动作（action）ID	int	动作（action）ID
type	动作类型	string	固定值：arcOff

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.3 描述：送丝

发送消息：

```
####000000a8{"function":"runTask","parameter":{"id":"171023527750479002","name":"push_wire","type":0,"actions":[{"id":"171023633117617300","type":"pushWire","pushWireTime":1000}]}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": "171023527750479002",
    "name": "push_wire",
    "type": 0,
    "actions": [
      {
        "id": "171023633117617300",
        "type": "pushWire",
        "pushWireTime": 1000
      }
    ]
  }
}
```

```
}
}
```

表 4-2-11 送丝动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	string	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
actions[0]	送丝	object	见下表
id	动作 (action) ID	int	动作 (action) ID
type	动作类型	string	固定值: pushWire
pushWireTime	送丝时间	int	送丝时间 (单位: ms)

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.4 描述：退丝

发送消息：

```
####000000bc{"function":"runTask","parameter":{"id":174852956135300000,"name":"tuisi","actions":[{"pullWireTime":1000,"id":174852979401000000,"type":"pullWire"}],"type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": 174852956135300000,
    "name": "tuisi",
    "actions": [
      {
        "pullWireTime": 1000,
        "id": 174852979401000000,
        "type": "pullWire"
      }
    ],
    "type": 0
  }
}
```

表 4-2-11 送丝动作-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值：runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID

name	任务名称	string	任务名称
type	任务类型	string	取值范围： 0：模拟任务，不起弧； 1：真实任务，起弧；
actions	动作列表	array	见下表
actions[0]	送丝	object	见下表
id	动作（action）ID	int	动作（action）ID
type	动作类型	string	固定值：pullWire
pullWireTime	送丝时间	int	退丝时间（单位：ms）

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.5 描述：直焊模版

发送消息：

```
####0000071d{"function":"runTask","parameter":{"id":"174494055348800000","name":"line_temp","actions":[{"children":[{"position":{"x":511.391,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-13.228,"J3":73.555,"J4":0,"J5":91.351,"J6":0,"TCP":"TCP","UCS":"Base"},"id":"174494065632400002","type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":"174494065632400001","type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSI
```

```
ope":0,"arcOnArcControl":0},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":"174494065632400003","type":"arcOn","arcOnClass":0},{ "children":[{"position":{"x":511.391,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-13.228,"J3":73.555,"J4":0,"J5":91.351,"J6":0,"TCP":"TCP","UCS":"Base"},"id":"174494065632500001","type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":"174494065632500000","type":"moveL"}, {"id":"174494065632500002","type":"arcOff"}]]}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": "174494055348800000",
    "name": "line_temp",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 511.391,
              "y": 0,
              "z": 457.068,
              "rx": 180,
              "ry": 1.866,
              "rz": 180,
              "J1": 0,
              "J2": -13.228,
              "J3": 73.555,
              "J4": 0,
              "J5": 91.351,
              "J6": 0,
              "TCP": "TCP",
              "UCS": "Base"
            }
          }
        ],
        "id": "174494065632400002",
        "type": "point"
      }
    ]
  }
}
```

```
    }  
  ],  
  "speed": 100,  
  "acceleration": 500,  
  "radius": 10,  
  "id": "174494065632400001",  
  "type": "moveL"  
},  
{  
  "arcOn": {  
    "arcOnMode": 0,  
    "arcOnCurrent": 100,  
    "arcOnAirPressure": 0,  
    "arcOnVoltage": 0,  
    "arcOnTime": 0,  
    "preflowTime": 0,  
    "arcOnPowerCorrection": 20,  
    "arcOnWireSpeed": 0,  
    "arcOnJobNumber": 1,  
    "arcOnInductance": 0,  
    "arcOnSlope": 0,  
    "arcOnArcControl": 0  
  },  
  "arcOff": {  
    "arcOffMode": 0,  
    "arcOffCurrent": 100,  
    "arcOffVoltage": 0,  
    "arcOffTime": 0,  
    "arcOffDistance": 0,  
    "enableArcOffDistance": true,  
    "postflowTime": 0,  
    "enableBurnback": 1,  
    "burnBackTime": 0,  
    "arcOffPowerCorrection": 20,  
    "arcOffWireSpeed": 0,  
    "arcOffJobNumber": 1,  
    "arcOffInductance": 0,  
    "arcOffSlope": 0,  
    "arcOffArcControl": 0  
  },  
  "arcSet": {  
    "arcSetMode": 0,  
    "arcSetCurrent": 100,  
    "arcSetAirPressure": 0,  
    "arcSetVoltage": 0,  
  }
```



```

    "arcSetPowerCorrection": 20,
    "arcSetWireSpeed": 0,
    "arcSetJobNumber": 1,
    "arcSetInductance": 0,
    "arcSetScanSpeed": 200,
    "arcSetScanLength": 2,
    "arcSetSpduty": 50,
    "arcSetSpfreq": 2000,
    "arcSetSwingLength": 0,
    "arcSetSwingFrequency": 0,
    "speed": 10,
    "enablePushWire": 0,
    "arcSetSlope": 0,
    "arcSetArcControl": 0
  },
  "weldProgram": {
    "isUseMemorySlot": 0,
    "programNumber": 1,
    "weldProcess": 0,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  },
  "id": "174494065632400003",
  "type": "arcOn",
  "arcOnClass": 0
},
{
  "children": [
    {
      "position": {
        "x": 511.391,
        "y": 0,
        "z": 457.068,
        "rx": 180,
        "ry": 1.866,
        "rz": 180,
        "J1": 0,
        "J2": -13.228,
        "J3": 73.555,
        "J4": 0,
        "J5": 91.351,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      }
    }
  ]
}

```

```

    }
    "id": "174494065632500001",
    "type": "point"
  }
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": "174494065632500000",
"type": "moveL"
},
{
  "id": "174494065632500002",
  "type": "arcOff"
}
],
"type": 0
}
}

```

表 4-2-12 直焊模版-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
actions[0]	直线	object	见 4.2.1.1 直线表

actions[1]	起弧	object	见 4.2.3.1 起弧表
actions[2]	直线	object	见 4.2.1.1 直线表
actions[3]	收弧	object	见 4.2.3.2 收弧表

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.6 描述：圆焊模版

发送消息:

```
#####00000883{"function":"runTask","parameter":{"id":174494125364500000,"name":"circle_temp","actions":[{"children":[{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494127176900000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174494127176900000,"type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174494127176900000,"type":"arcOn","arcOnClass":0},{children":[{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494127176900000,"type":"point"}]}
```

```
94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494127176900000,"type":"point"},{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494127176900000,"type":"point"}], "speed":100,"acceleration":500,"radius":10,"movecType":1,"radian":1,"fixedPosture":0,"id":174494127176900000,"type":"moveC"}, {"id":174494127176900000,"type":"arcOff"}], "type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": 174494125364500000,
    "name": "circle_temp",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 574.71,
              "y": 0,
              "z": 457.068,
              "rx": 180,
              "ry": 1.866,
              "rz": 180,
              "J1": 0,
              "J2": -23.588,
              "J3": 60.23,
              "J4": 0,
              "J5": 94.317,
              "J6": 0,
              "TCP": "TCP",
              "UCS": "Base"
            },
            "enableVariable": 0,
            "variableSymbol": "p1",
            "id": 174494127176900000,
            "type": "point"
          }
        ],
        "speed": 100,
        "acceleration": 500,
        "radius": 10,
        "id": 174494127176900000,
```

```
"type": "moveL"  
},  
{  
  "arcOn": {  
    "arcOnMode": 0,  
    "arcOnCurrent": 100,  
    "arcOnAirPressure": 0,  
    "arcOnVoltage": 0,  
    "arcOnTime": 0,  
    "preflowTime": 0,  
    "arcOnPowerCorrection": 20,  
    "arcOnWireSpeed": 0,  
    "arcOnJobNumber": 1,  
    "arcOnInductance": 0,  
    "arcOnSlope": 0,  
    "arcOnArcControl": 0  
  },  
  "arcOff": {  
    "arcOffMode": 0,  
    "arcOffCurrent": 100,  
    "arcOffVoltage": 0,  
    "arcOffTime": 0,  
    "arcOffDistance": 0,  
    "enableArcOffDistance": true,  
    "postflowTime": 0,  
    "enableBurnback": 1,  
    "burnBackTime": 0,  
    "arcOffPowerCorrection": 20,  
    "arcOffWireSpeed": 0,  
    "arcOffJobNumber": 1,  
    "arcOffInductance": 0,  
    "arcOffSlope": 0,  
    "arcOffArcControl": 0  
  },  
  "arcSet": {  
    "arcSetMode": 0,  
    "arcSetCurrent": 100,  
    "arcSetAirPressure": 0,  
    "arcSetVoltage": 0,  
    "arcSetPowerCorrection": 20,  
    "arcSetWireSpeed": 0,  
    "arcSetJobNumber": 1,  
    "arcSetInductance": 0,  
    "arcSetScanSpeed": 200,  
    "arcSetScanLength": 2,  
  }  
}
```

```

        "arcSetSpduty": 50,
        "arcSetSpfreq": 2000,
        "arcSetSwingLength": 0,
        "arcSetSwingFrequency": 0,
        "speed": 10,
        "enablePushWire": 0,
        "arcSetSlope": 0,
        "arcSetArcControl": 0
    },
    "weldProgram": {
        "isUseMemorySlot": 0,
        "programNumber": 1,
        "weldProcess": 0,
        "weldWireType": 0,
        "weldWireSize": 0,
        "weldGasType": 0
    }
    "id": 174494127176900000,
    "type": "arcOn",
    "arcOnClass": 0
},
{
    "children": [
        {
            "position": {
                "x": 574.71,
                "y": 0,
                "z": 457.068,
                "rx": 180,
                "ry": 1.866,
                "rz": 180,
                "J1": 0,
                "J2": -23.588,
                "J3": 60.23,
                "J4": 0,
                "J5": 94.317,
                "J6": 0,
                "TCP": "TCP",
                "UCS": "Base"
            },
            "enableVariable": 0,
            "variableSymbol": "p1",
            "id": 174494127176900000,
            "type": "point"
        },
    ],
},

```

```

    {
      "position": {
        "x": 574.71,
        "y": 0,
        "z": 457.068,
        "rx": 180,
        "ry": 1.866,
        "rz": 180,
        "J1": 0,
        "J2": -23.588,
        "J3": 60.23,
        "J4": 0,
        "J5": 94.317,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174494127176900000,
      "type": "point"
    }
  ],
  "speed": 100,
  "acceleration": 500,
  "radius": 10,
  "movecType": 1,
  "radian": 1,
  "fixedPosture": 0
  "id": 174494127176900000,
  "type": "moveC"
},
{
  "id": 174494127176900000,
  "type": "arcOff"
}
],
"type": 0
}
}

```

表 4-2-13 圆焊模版-消息字段说明

参数	名称	数据类型	内容
----	----	------	----

function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
actions[0]	直线	object	见 4.2.1.1 直线表
actions[1]	起弧	object	见 4.2.3.1 起弧表
actions[2]	圆弧	object	见 4.2.1.3 圆弧表
actions[3]	收弧	object	见 4.2.3.2 收弧表
weldProgram	焊接程序	object	见 4.1.20 (仅米勒 ModBus TCP/IP 模式)

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.7 描述: 摆焊模版

发送消息：

```
####0000086a{"function":"runTask","parameter":{"id":174494196557600000,"name":"swing_temp","actions":[{"children":[{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494198453000000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174494198453000000,"type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174494198453000000,"type":"arcOn","arcOnClass":0},{"shape":"zigzag","amplitude":2,"frequency":2,"dwellEnable":0,"dwellLeft":500,"dwellRight":500,"weaveFrameType":0,"centreRise":0,"elevation":0,"azimuth":0,"id":174494198453000000,"type":"weaveOn"},{"children":[{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494198453000000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174494198453000000,"type":"moveL"},{"id":174494198453000000,"type":"weaveOff"},{"id":174494198453100000,"type":"arcOff"}],"type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": 174494196557600000,
    "name": "swing_temp",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 574.71,
              "y": 0,
              "z": 457.068,
              "rx": 180,
              "ry": 1.866,
```

```

    "rz": 180,
    "J1": 0,
    "J2": -23.588,
    "J3": 60.23,
    "J4": 0,
    "J5": 94.317,
    "J6": 0,
    "TCP": "TCP",
    "UCS": "Base"
  },
  "enableVariable": 0,
  "variableSymbol": "p1",

  "id": 174494198453000000,
  "type": "point"
}
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": 174494198453000000,
"type": "moveL"
},
{
  "arcOn": {
    "arcOnMode": 0,
    "arcOnCurrent": 100,
    "arcOnAirPressure": 0,
    "arcOnVoltage": 0,
    "arcOnTime": 0,
    "preflowTime": 0,
    "arcOnPowerCorrection": 20,
    "arcOnWireSpeed": 0,
    "arcOnJobNumber": 1,
    "arcOnInductance": 0,
    "arcOnSlope": 0,
    "arcOnArcControl": 0
  },
  "arcOff": {
    "arcOffMode": 0,
    "arcOffCurrent": 100,
    "arcOffVoltage": 0,
    "arcOffTime": 0,
    "arcOffDistance": 0,
    "enableArcOffDistance": true,

```

```

    "postflowTime": 0,
    "enableBurnback": 1,
    "burnBackTime": 0,
    "arcOffPowerCorrection": 20,
    "arcOffWireSpeed": 0,
    "arcOffJobNumber": 1,
    "arcOffInductance": 0,
    "arcOffSlope": 0,
    "arcOffArcControl": 0
  },
  "arcSet": {
    "arcSetMode": 0,
    "arcSetCurrent": 100,
    "arcSetAirPressure": 0,
    "arcSetVoltage": 0,
    "arcSetPowerCorrection": 20,
    "arcSetWireSpeed": 0,
    "arcSetJobNumber": 1,
    "arcSetInductance": 0,
    "arcSetScanSpeed": 200,
    "arcSetScanLength": 2,
    "arcSetSpduty": 50,
    "arcSetSpfreq": 2000,
    "arcSetSwingLength": 0,
    "arcSetSwingFrequency": 0,
    "speed": 10,
    "enablePushWire": 0,
    "arcSetSlope": 0,
    "arcSetArcControl": 0
  },
  "weldProgram": {
    "isUseMemorySlot": 0,
    "programNumber": 1,
    "weldProcess": 0,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  },
  "id": 174494198453000000,
  "type": "arcOn",
  "arcOnClass": 0
},
{
  "shape": "zigzag",
  "amplitude": 2,

```

```

    "frequency": 2,
    "dwellEnable": 0,
    "dwellLeft": 500,
    "dwellRight": 500,
    "weaveFrameType": 0,
    "centreRise": 0,
    "elevation": 0,
    "azimuth": 0,
    "id": 174494198453000000,
    "type": "weaveOn"
  },
  {
    "children": [
      {
        "position": {
          "x": 574.71,
          "y": 0,
          "z": 457.068,
          "rx": 180,
          "ry": 1.866,
          "rz": 180,
          "J1": 0,
          "J2": -23.588,
          "J3": 60.23,
          "J4": 0,
          "J5": 94.317,
          "J6": 0,
          "TCP": "TCP",
          "UCS": "Base"
        },
        "enableVariable": 0,
        "variableSymbol": "p1",

        "id": 174494198453000000,
        "type": "point"
      }
    ],
    "speed": 100,
    "acceleration": 500,
    "radius": 10,
    "id": 174494198453000000,
    "type": "moveL"
  },
  {

```

```

        "id": 174494198453000000,
        "type": "weaveOff"
    },
    {
        "id": 174494198453100000,
        "type": "arcOff"
    }
],
"type": 0
}
}

```

表 4-2-14 摆焊模版-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
actions[0]	直线	object	见 4.2.1.1 直线表
actions[1]	起弧	object	见 4.2.3.1 起弧表
actions[2]	摆动开始	object	见 4.2.1.4 摆动表

actions[3]	摆动结束	object	见 4.2.1.4 摆动表
actions[4]	收弧	object	见 4.2.3.2 收弧表

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.8 描述：拼焊模版

发送消息：

```
####000009d4{"function":"runTask","parameter":{"id":174494280401800000,"name":"splice_temp","actions":[{"children":[{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494281584100000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174494281584100000,"type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0,"id":174494281584100000,"type":"arcOn","arcOnClass":0},"children":[{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494281584100000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174494281584100000,"type":"moveL"},{"children":[{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174494281584100000,"t
```

```

type:"point"},{"position":{"x":574.71,"y":0,"z":457.068,"rx":180,"ry":1.866,"rz":180,"J1":0,"J
2":-23.588,"J3":60.23,"J4":0,"J5":94.317,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariabl
e":0,"variableSymbol":"p1","id":174494281584100000,"type":"point"}],"speed":100,"accel
eration":500,"radius":10,"movecType":1,"radian":1,"fixedPosture":0,"id":17449428158410
0000,"type":"moveC"},{"id":174494281584200000,"type":"arcOff"}],"type":0}}$$$$

```

消息体 JSON 格式化:

```

{
  "function": "runTask",
  "parameter": {
    "id": 174494280401800000,
    "name": "splice_temp",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 574.71,
              "y": 0,
              "z": 457.068,
              "rx": 180,
              "ry": 1.866,
              "rz": 180,
              "J1": 0,
              "J2": -23.588,
              "J3": 60.23,
              "J4": 0,
              "J5": 94.317,
              "J6": 0,
              "TCP": "TCP",
              "UCS": "Base"
            },
            "enableVariable": 0,
            "variableSymbol": "p1",
            "id": 174494281584100000,
            "type": "point"
          }
        ],
        "speed": 100,
        "acceleration": 500,
        "radius": 10,

```

```
"id": 174494281584100000,
"type": "moveL"
},
{
  "arcOn": {
    "arcOnMode": 0,
    "arcOnCurrent": 100,
    "arcOnAirPressure": 0,
    "arcOnVoltage": 0,
    "arcOnTime": 0,
    "preflowTime": 0,
    "arcOnPowerCorrection": 20,
    "arcOnWireSpeed": 0,
    "arcOnJobNumber": 1,
    "arcOnInductance": 0,
    "arcOnSlope": 0,
    "arcOnArcControl": 0
  },
  "arcOff": {
    "arcOffMode": 0,
    "arcOffCurrent": 100,
    "arcOffVoltage": 0,
    "arcOffTime": 0,
    "arcOffDistance": 0,
    "enableArcOffDistance": true,
    "postflowTime": 0,
    "enableBurnback": 1,
    "burnBackTime": 0,
    "arcOffPowerCorrection": 20,
    "arcOffWireSpeed": 0,
    "arcOffJobNumber": 1,
    "arcOffInductance": 0,
    "arcOffSlope": 0,
    "arcOffArcControl": 0
  },
  "arcSet": {
    "arcSetMode": 0,
    "arcSetCurrent": 100,
    "arcSetAirPressure": 0,
    "arcSetVoltage": 0,
    "arcSetPowerCorrection": 20,
    "arcSetWireSpeed": 0,
    "arcSetJobNumber": 1,
    "arcSetInductance": 0,
    "arcSetScanSpeed": 200,
```



```

    "arcSetScanLength": 2,
    "arcSetSpduty": 50,
    "arcSetSpfreq": 2000,
    "arcSetSwingLength": 0,
    "arcSetSwingFrequency": 0,
    "speed": 10,
    "enablePushWire": 0,
    "arcSetSlope": 0,
    "arcSetArcControl": 0
  },
  "weldProgram": {
    "isUseMemorySlot": 0,
    "programNumber": 1,
    "weldProcess": 0,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  },
  "id": 174494281584100000,
  "type": "arcOn",
  "arcOnClass": 0
},
{
  "children": [
    {
      "position": {
        "x": 574.71,
        "y": 0,
        "z": 457.068,
        "rx": 180,
        "ry": 1.866,
        "rz": 180,
        "J1": 0,
        "J2": -23.588,
        "J3": 60.23,
        "J4": 0,
        "J5": 94.317,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174494281584100000,
      "type": "point"
    }
  ]
}

```

```

    }
  ],
  "speed": 100,
  "acceleration": 500,
  "radius": 10,
  "id": 174494281584100000,
  "type": "moveL"
},
{
  "children": [
    {
      "position": {
        "x": 574.71,
        "y": 0,
        "z": 457.068,
        "rx": 180,
        "ry": 1.866,
        "rz": 180,
        "J1": 0,
        "J2": -23.588,
        "J3": 60.23,
        "J4": 0,
        "J5": 94.317,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174494281584100000,
      "type": "point"
    },
    {
      "position": {
        "x": 574.71,
        "y": 0,
        "z": 457.068,
        "rx": 180,
        "ry": 1.866,
        "rz": 180,
        "J1": 0,
        "J2": -23.588,
        "J3": 60.23,
        "J4": 0,
        "J5": 94.317,

```

```

        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174494281584100000,
    "type": "point"
  }
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"movecType": 1,
"radian": 1,
"fixedPosture": 0,
"id": 174494281584100000,
"type": "moveC"
},
{
  "id": 174494281584200000,
  "type": "arcOff"
}
],
"type": 0
}
}

```

表 4-2-15 拼焊模版-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围:

			0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表
children[0]	直线	object	见 4.2.1.1 直线表
children[1]	起弧	object	见 4.3.3.1 起弧表
children[2]	直线	object	见 4.2.1.1 直线表
children[3]	圆弧	object	见 4.2.1.3 圆弧表
children[1]	收弧	object	见 4.3.3.2 收弧表

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.9 描述: 多层多道模版

(多层多道自定义) 发送消息:

```
####000013fe{"function":"runTask","parameter":{"id":174364088670100000,"name":1,"actions":[{"configType":1,"beginPass":[1,1],"endPass":[2,2],"layerConfig":{"headPadding":0,"tailPadding":0,"layerPasses":1,"offsets":[0,0,0],"arcCraft":{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":10,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":50,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0,"arcOnControl":25},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":10,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":50,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0,"arcOffControl":25},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetA
```

```

irPressure":0,"arcSetVoltage":10,"arcSetPowerCorrection":20,"arcSetWireSpeed":50,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0,"arcSetControl":25},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174364089723900000,"type":"arcOn","arcOnClass":0},"weaveCraft":"","headPadding":0,"tailPadding":0,"layerPasses":2,"offsets":[1,1,1,2,2,2],"arcCraft":{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":10,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":50,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0,"arcOnControl":25},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":10,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":50,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0,"arcOffControl":25},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":10,"arcSetPowerCorrection":20,"arcSetWireSpeed":50,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0,"arcSetControl":25},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174364089723900000,"type":"arcOn","arcOnClass":0},"weaveCraft":"","multiPassMultiLayerOffsetId":174364089723900000,"otherData":{"beginPass":[1,1],"endPass":[1,1],"layerConfig":[{"headPadding":0,"tailPadding":0,"arcCraft":"","weaveCraft":"","layerPasses":1,"offsets":[0,0,0]}],"id":174364089723800000,"type":"multiPassMultiLayerOn","children":[{"children":[{"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174401064642100000,"type":"point"},"speed":100,"acceleration":500,"radius":10,"id":174401064642100000,"type":"moveL"},"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":10,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":50,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0,"arcOnControl":25},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":10,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":50,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0,"arcOffControl":25},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":10,"arcSetPowerCorrection":20,"arcSetWireSpeed":50,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0,"arcSetControl":25},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174364089723900000,"type":"arcOn","arcOnClass":0},"children":[{"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174364089723900000,"type":"point"},"speed":100,"acceleration":500,"radius":10,"id":174364089723900000,"type":"moveL"},"id":174364089723900000,"type":"arcOff"},"children":[{"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVari

```

```
able":0,"variableSymbol":"p1","id":174364089723900000,"type":"point"},"speed":100,"acceleration":500,"radius":10,"id":174364089723900000,"type":"moveL"}},{ "multiPassMultiLayerOnId":174364089723800000,"id":174364089723900000,"type":"multiPassMultiLayerOff"}], "type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": 174364088670100000,
    "name": 1,
    "actions": [
      {
        "configType": 1,
        "beginPass": [
          1,
          1
        ],
        "endPass": [
          2,
          2
        ],
        "layerConfig": [
          {
            "headPadding": 0,
            "tailPadding": 0,
            "layerPasses": 1,
            "offsets": [
              0,
              0,
              0
            ],
            "arcCraft": {
              "arcOn": {
                "arcOnMode": 0,
                "arcOnCurrent": 100,
                "arcOnAirPressure": 0,
                "arcOnVoltage": 10,
                "arcOnTime": 0,
                "preflowTime": 0,
                "arcOnPowerCorrection": 20,
                "arcOnWireSpeed": 50,
                "arcOnJobNumber": 1,
```

```
"arcOnInductance": 0,  
"arcOnSlope": 0,  
"arcOnArcControl": 0,  
"arcOnControl": 25  
},  
"arcOff": {  
  "arcOffMode": 0,  
  "arcOffCurrent": 100,  
  "arcOffVoltage": 10,  
  "arcOffTime": 0,  
  "arcOffDistance": 0,  
  "enableArcOffDistance": true,  
  "postflowTime": 0,  
  "enableBurnback": 1,  
  "burnBackTime": 0,  
  "arcOffPowerCorrection": 20,  
  "arcOffWireSpeed": 50,  
  "arcOffJobNumber": 1,  
  "arcOffInductance": 0,  
  "arcOffSlope": 0,  
  "arcOffArcControl": 0,  
  "arcOffControl": 25  
},  
"arcSet": {  
  "arcSetMode": 0,  
  "arcSetCurrent": 100,  
  "arcSetAirPressure": 0,  
  "arcSetVoltage": 10,  
  "arcSetPowerCorrection": 20,  
  "arcSetWireSpeed": 50,  
  "arcSetJobNumber": 1,  
  "arcSetInductance": 0,  
  "arcSetScanSpeed": 200,  
  "arcSetScanLength": 2,  
  "arcSetSpduty": 50,  
  "arcSetSpfreq": 2000,  
  "arcSetSwingLength": 0,  
  "arcSetSwingFrequency": 0,  
  "speed": 10,  
  "enablePushWire": 0,  
  "arcSetSlope": 0,  
  "arcSetArcControl": 0,  
  "arcSetControl": 25  
},  
"weldProgram": {
```

```
        "isUseMemorySlot": 0,
        "programNumber": 1,
        "weldProcess": 0,
        "weldWireType": 0,
        "weldWireSize": 0,
        "weldGasType": 0
    },
    "id": 174364089723900000,
    "type": "arcOn",
    "arcOnClass": 0
},
"weaveCraft": ""
},
{
    "headPadding": 0,
    "tailPadding": 0,
    "layerPasses": 2,
    "offsets": [
        1,
        1,
        1,
        2,
        2,
        2
    ],
    "arcCraft": {
        "arcOn": {
            "arcOnMode": 0,
            "arcOnCurrent": 100,
            "arcOnAirPressure": 0,
            "arcOnVoltage": 10,
            "arcOnTime": 0,
            "preflowTime": 0,
            "arcOnPowerCorrection": 20,
            "arcOnWireSpeed": 50,
            "arcOnJobNumber": 1,
            "arcOnInductance": 0,
            "arcOnSlope": 0,
            "arcOnArcControl": 0,
            "arcOnControl": 25
        },
        "arcOff": {
            "arcOffMode": 0,
            "arcOffCurrent": 100,
            "arcOffVoltage": 10,
```



```
"arcOffTime": 0,
"arcOffDistance": 0,
"enableArcOffDistance": true,
"postflowTime": 0,
"enableBurnback": 1,
"burnBackTime": 0,
"arcOffPowerCorrection": 20,
"arcOffWireSpeed": 50,
"arcOffJobNumber": 1,
"arcOffInductance": 0,
"arcOffSlope": 0,
"arcOffArcControl": 0,
"arcOffControl": 25
},
"arcSet": {
  "arcSetMode": 0,
  "arcSetCurrent": 100,
  "arcSetAirPressure": 0,
  "arcSetVoltage": 10,
  "arcSetPowerCorrection": 20,
  "arcSetWireSpeed": 50,
  "arcSetJobNumber": 1,
  "arcSetInductance": 0,
  "arcSetScanSpeed": 200,
  "arcSetScanLength": 2,
  "arcSetSpduty": 50,
  "arcSetSpfreq": 2000,
  "arcSetSwingLength": 0,
  "arcSetSwingFrequency": 0,
  "speed": 10,
  "enablePushWire": 0,
  "arcSetSlope": 0,
  "arcSetArcControl": 0,
  "arcSetControl": 25
},
"weldProgram": {
  "isUseMemorySlot": 0,
  "programNumber": 1,
  "weldProcess": 0,
  "weldWireType": 0,
  "weldWireSize": 0,
  "weldGasType": 0
},
"id": 174364089723900000,
"type": "arcOn",
```

```
        "arcOnClass": 0
      },
      "weaveCraft": ""
    }
  ],
  "multiPassMultiLayerOffId": 174364089723900000,
  "otherData": {
    "beginPass": [
      1,
      1
    ],
    "endPass": [
      1,
      1
    ],
    "layerConfig": [
      {
        "headPadding": 0,
        "tailPadding": 0,
        "arcCraft": "",
        "weaveCraft": "",
        "layerPasses": 1,
        "offsets": [
          0,
          0,
          0
        ]
      }
    ]
  },
  "id": 174364089723800000,
  "type": "multiPassMultiLayerOn",
  "children": [
    {
      "children": [
        {
          "position": {
            "x": 0,
            "y": 0,
            "z": 966.5,
            "rx": 0,
            "ry": 0,
            "rz": 0,
            "J1": 0,
            "J2": 0,
```

```

    "J3": 0,
    "J4": 0,
    "J5": 0,
    "J6": 0,
    "TCP": "TCP",
    "UCS": "Base"
  },
  "enableVariable": 0,
  "variableSymbol": "p1",
  "id": 174401064642100000,
  "type": "point"
}
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": 174401064642100000,
"type": "moveL"
},
{
  "arcOn": {
    "arcOnMode": 0,
    "arcOnCurrent": 100,
    "arcOnAirPressure": 0,
    "arcOnVoltage": 10,
    "arcOnTime": 0,
    "preflowTime": 0,
    "arcOnPowerCorrection": 20,
    "arcOnWireSpeed": 50,
    "arcOnJobNumber": 1,
    "arcOnInductance": 0,
    "arcOnSlope": 0,
    "arcOnArcControl": 0,
    "arcOnControl": 25
  },
  "arcOff": {
    "arcOffMode": 0,
    "arcOffCurrent": 100,
    "arcOffVoltage": 10,
    "arcOffTime": 0,
    "arcOffDistance": 0,
    "enableArcOffDistance": true,
    "postflowTime": 0,
    "enableBurnback": 1,
    "burnBackTime": 0,

```

```

        "arcOffPowerCorrection": 20,
        "arcOffWireSpeed": 50,
        "arcOffJobNumber": 1,
        "arcOffInductance": 0,
        "arcOffSlope": 0,
        "arcOffArcControl": 0,
        "arcOffControl": 25
    },
    "arcSet": {
        "arcSetMode": 0,
        "arcSetCurrent": 100,
        "arcSetAirPressure": 0,
        "arcSetVoltage": 10,
        "arcSetPowerCorrection": 20,
        "arcSetWireSpeed": 50,
        "arcSetJobNumber": 1,
        "arcSetInductance": 0,
        "arcSetScanSpeed": 200,
        "arcSetScanLength": 2,
        "arcSetSpduty": 50,
        "arcSetSpfreq": 2000,
        "arcSetSwingLength": 0,
        "arcSetSwingFrequency": 0,
        "speed": 10,
        "enablePushWire": 0,
        "arcSetSlope": 0,
        "arcSetArcControl": 0,
        "arcSetControl": 25
    },
    "weldProgram": {
        "isUseMemorySlot": 0,
        "programNumber": 1,
        "weldProcess": 0,
        "weldWireType": 0,
        "weldWireSize": 0,
        "weldGasType": 0
    },
    "id": 174364089723900000,
    "type": "arcOn",
    "arcOnClass": 0
},
{
    "children": [
        {
            "position": {

```

```
        "x": 0,
        "y": 0,
        "z": 966.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174364089723900000,
    "type": "point"
  }
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": 174364089723900000,
"type": "moveL"
},
{
  "id": 174364089723900000,
  "type": "arcOff"
},
{
  "children": [
    {
      "position": {
        "x": 0,
        "y": 0,
        "z": 966.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
```

```

        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174364089723900000,
    "type": "point"
  }
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": 174364089723900000,
"type": "moveL"
}
]
},
{
  "multiPassMultiLayerOnId": 174364089723800000,
  "id": 174364089723900000,
  "type": "multiPassMultiLayerOff"
}
],
"type": 0
}
}

```

（多层多道自适应）发送消息：

```

####0000118b{"function":"runTask","parameter":{"id":174364088670100000,"name":1,"
actions":[{"children":[{"children":[{"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"
J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSy
mbol":"p1","id":174401064642100000,"type":"point"}],"speed":100,"acceleration":500,"ra
dius":10,"id":174401064642100000,"type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurr
ent":100,"arcOnAirPressure":0,"arcOnVoltage":10,"arcOnTime":0,"preflowTime":0,"arcOnP
owerCorrection":20,"arcOnWireSpeed":50,"arcOnJobNumber":1,"arcOnInductance":0,"arc
OnSlope":0,"arcOnArcControl":0,"arcOnControl":25},"arcOff":{"arcOffMode":0,"arcOffCurre
nt":100,"arcOffVoltage":10,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,
"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arc
OffWireSpeed":50,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArc
Control":0,"arcOffControl":25},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPre

```

```
ssure":0,"arcSetVoltage":10,"arcSetPowerCorrection":20,"arcSetWireSpeed":50,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0,"arcSetControl":25},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174364089723900000,"type":"arcOn","arcOnClass":0},{ "children":[{"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174364089723900000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174364089723900000,"type":"moveL"}, {"id":174364089723900000,"type":"arcOff"}, {"children":[{"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174364089723900000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174364089723900000,"type":"moveL"}],"weldType":0,"weldThickness":10,"configType":0,"layerThickness":10,"beginPass":[1,1],"endPass":[1,1],"directPoint":1,"layerConfig":{"headPadding":0,"tailPadding":0,"arcCraft":{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":10,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":50,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0,"arcOnControl":25},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":10,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":50,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0,"arcOffControl":25},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":10,"arcSetPowerCorrection":20,"arcSetWireSpeed":50,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0,"arcSetControl":25},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174364089723900000,"type":"arcOn","arcOnClass":0},"weaveCraft":"","layerPasses":1,"offsets":[0,0,0]}],"multiPassMultiLayerOffId":174364089723900000,"id":174364089723800000,"type":"multiPassMultiLayerOn","auxiliaryPoints":[{"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174364089723800000,"type":"point","directPoint":false}, {"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174364089723900000,"type":"point","directPoint":true}, {"position":{"x":0,"y":0,"z":966.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174364089723900000,"type":"point","directPoint":false}]},"multiPassMultiLayerOnId":174364089723800000,"id":174364089723900000,"type":"multiPassMultiLayerOff"}],"type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
```

```
"parameter": {
  "id": 174364088670100000,
  "name": 1,
  "actions": [
    {
      "children": [
        {
          "children": [
            {
              "position": {
                "x": 0,
                "y": 0,
                "z": 966.5,
                "rx": 0,
                "ry": 0,
                "rz": 0,
                "J1": 0,
                "J2": 0,
                "J3": 0,
                "J4": 0,
                "J5": 0,
                "J6": 0,
                "TCP": "TCP",
                "UCS": "Base"
              },
              "enableVariable": 0,
              "variableSymbol": "p1",
              "id": 174401064642100000,
              "type": "point"
            }
          ],
          "speed": 100,
          "acceleration": 500,
          "radius": 10,
          "id": 174401064642100000,
          "type": "moveL"
        },
        {
          "arcOn": {
            "arcOnMode": 0,
            "arcOnCurrent": 100,
            "arcOnAirPressure": 0,
            "arcOnVoltage": 10,
            "arcOnTime": 0,
            "preflowTime": 0,
```



```
"arcOnPowerCorrection": 20,
"arcOnWireSpeed": 50,
"arcOnJobNumber": 1,
"arcOnInductance": 0,
"arcOnSlope": 0,
"arcOnArcControl": 0,
"arcOnControl": 25
},
"arcOff": {
  "arcOffMode": 0,
  "arcOffCurrent": 100,
  "arcOffVoltage": 10,
  "arcOffTime": 0,
  "arcOffDistance": 0,
  "enableArcOffDistance": true,
  "postflowTime": 0,
  "enableBurnback": 1,
  "burnBackTime": 0,
  "arcOffPowerCorrection": 20,
  "arcOffWireSpeed": 50,
  "arcOffJobNumber": 1,
  "arcOffInductance": 0,
  "arcOffSlope": 0,
  "arcOffArcControl": 0,
  "arcOffControl": 25
},
"arcSet": {
  "arcSetMode": 0,
  "arcSetCurrent": 100,
  "arcSetAirPressure": 0,
  "arcSetVoltage": 10,
  "arcSetPowerCorrection": 20,
  "arcSetWireSpeed": 50,
  "arcSetJobNumber": 1,
  "arcSetInductance": 0,
  "arcSetScanSpeed": 200,
  "arcSetScanLength": 2,
  "arcSetSpduty": 50,
  "arcSetSpfreq": 2000,
  "arcSetSwingLength": 0,
  "arcSetSwingFrequency": 0,
  "speed": 10,
  "enablePushWire": 0,
  "arcSetSlope": 0,
  "arcSetArcControl": 0,
```

```

    "arcSetControl": 25
  },
  "weldProgram": {
    "isUseMemorySlot": 0,
    "programNumber": 1,
    "weldProcess": 0,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  },
  "id": 174364089723900000,
  "type": "arcOn",
  "arcOnClass": 0
},
{
  "children": [
    {
      "position": {
        "x": 0,
        "y": 0,
        "z": 966.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174364089723900000,
      "type": "point"
    }
  ],
  "speed": 100,
  "acceleration": 500,
  "radius": 10,
  "id": 174364089723900000,
  "type": "moveL"
},

```

```

    {
      "id": 174364089723900000,
      "type": "arcOff"
    },
    {
      "children": [
        {
          "position": {
            "x": 0,
            "y": 0,
            "z": 966.5,
            "rx": 0,
            "ry": 0,
            "rz": 0,
            "J1": 0,
            "J2": 0,
            "J3": 0,
            "J4": 0,
            "J5": 0,
            "J6": 0,
            "TCP": "TCP",
            "UCS": "Base"
          },
          "enableVariable": 0,
          "variableSymbol": "p1",
          "id": 174364089723900000,
          "type": "point"
        }
      ],
      "speed": 100,
      "acceleration": 500,
      "radius": 10,
      "id": 174364089723900000,
      "type": "moveL"
    }
  ],
  "weldType": 0,
  "weldThickness": 10,
  "configType": 0,
  "layerThickness": 10,
  "beginPass": [
    1,
    1
  ],
  "endPass": [

```

```

        1,
        1
    ],
    "directPoint": 1,
    "layerConfig": [
        {
            "headPadding": 0,
            "tailPadding": 0,
            "arcCraft": {
                "arcOn": {
                    "arcOnMode": 0,
                    "arcOnCurrent": 100,
                    "arcOnAirPressure": 0,
                    "arcOnVoltage": 10,
                    "arcOnTime": 0,
                    "preflowTime": 0,
                    "arcOnPowerCorrection": 20,
                    "arcOnWireSpeed": 50,
                    "arcOnJobNumber": 1,
                    "arcOnInductance": 0,
                    "arcOnSlope": 0,
                    "arcOnArcControl": 0,
                    "arcOnControl": 25
                },
                "arcOff": {
                    "arcOffMode": 0,
                    "arcOffCurrent": 100,
                    "arcOffVoltage": 10,
                    "arcOffTime": 0,
                    "arcOffDistance": 0,
                    "enableArcOffDistance": true,
                    "postflowTime": 0,
                    "enableBurnback": 1,
                    "burnBackTime": 0,
                    "arcOffPowerCorrection": 20,
                    "arcOffWireSpeed": 50,
                    "arcOffJobNumber": 1,
                    "arcOffInductance": 0,
                    "arcOffSlope": 0,
                    "arcOffArcControl": 0,
                    "arcOffControl": 25
                },
                "arcSet": {
                    "arcSetMode": 0,
                    "arcSetCurrent": 100,

```

```

    "arcSetAirPressure": 0,
    "arcSetVoltage": 10,
    "arcSetPowerCorrection": 20,
    "arcSetWireSpeed": 50,
    "arcSetJobNumber": 1,
    "arcSetInductance": 0,
    "arcSetScanSpeed": 200,
    "arcSetScanLength": 2,
    "arcSetSpduty": 50,
    "arcSetSpfreq": 2000,
    "arcSetSwingLength": 0,
    "arcSetSwingFrequency": 0,
    "speed": 10,
    "enablePushWire": 0,
    "arcSetSlope": 0,
    "arcSetArcControl": 0,
    "arcSetControl": 25
  },
  "weldProgram": {
    "isUseMemorySlot": 0,
    "programNumber": 1,
    "weldProcess": 0,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  },
  "id": 174364089723900000,
  "type": "arcOn",
  "arcOnClass": 0
},
"weaveCraft": "",
"layerPasses": 1,
"offsets": [
  0,
  0,
  0
]
}
],
"multiPassMultiLayerOffId": 174364089723900000,
"id": 174364089723800000,
"type": "multiPassMultiLayerOn",
"auxiliaryPoints": [
  {
    "position": {

```

```
        "x": 0,
        "y": 0,
        "z": 966.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174364089723800000,
    "type": "point",
    "directPoint": false
},
{
    "position": {
        "x": 0,
        "y": 0,
        "z": 966.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174364089723900000,
    "type": "point",
    "directPoint": true
},
{
```

```

    "position": {
      "x": 0,
      "y": 0,
      "z": 966.5,
      "rx": 0,
      "ry": 0,
      "rz": 0,
      "J1": 0,
      "J2": 0,
      "J3": 0,
      "J4": 0,
      "J5": 0,
      "J6": 0,
      "TCP": "TCP",
      "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174364089723900000,
    "type": "point",
    "directPoint": false
  }
]
},
{
  "multiPassMultiLayerOnId": 174364089723800000,
  "id": 174364089723900000,
  "type": "multiPassMultiLayerOff"
}
],
"type": 0
}
}

```

表 4-2-16 多层多道模版-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表

id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围： 0: 模拟任务，不起弧； 1: 真实任务，起弧；
actions	动作列表	array	见下表
actions[0].children[0]	直线	object	见 4.3.1.1 直线表
actions[0].children[1]	起弧	object	见 4.3.3.1 起弧表
actions[0].children[2]	直线	object	见 4.3.1.1 直线表
actions[0].children[3]	收弧	object	见 4.3.3.2 收弧表
actions[0].children[4]	直线	object	见 4.3.1.1 直线表
actions[0].weldType	焊接类型	int	固定值： 0: 直角焊缝（一个辅助点） 1: 对接焊缝（两个辅助点）
actions[0].weldThickness	焊脚高度	int	焊脚高度（单位：mm）
actions[0].layerThickness	层厚度	int	层厚度（单位：mm）
actions[0].beginPass	起始层道	array	大小为 2 的数组，第一个表示要焊的起始层、第二个表示要焊的起始道

actions[0].endPass	结束层道		array	大小为 2 的数组, 第一个表示要焊的结束层、第二个表示要焊的结束道
actions[0].directPoint	抬升方向		int	抬升方向点位的下标
actions[0].configType	焊接模式		int	0: 自适应模式 1: 专家模式
actions[0].layerConfig	layerConfig[0]	headPadding	int	头缩进
		arcCraft	object	起弧焊接配置参数
		tailPadding	int	尾部缩进
		weaveCraft	object	摆动配置
		layerPasses	int	每层的道数
		offsets	array	自定义模式下的偏移量, 分别对应 x、y、z 坐标的偏移量
actions[0].auxiliaryPoints	确定焊缝形状的点位		array	里面的三个点位确定了焊缝的形状: V 型、单 V 或者直角
actions[0].id	动作 id		string	动作 id
actions[0].type	动作类型		string	动作类型, 固定值: multiPassMultiLayerOn
actions[1]	多层多道结束		object	动作类型, 固定值: multiPassMultiLayer

			Off
--	--	--	-----

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.10 描述：间断焊模版

发送消息：

```
####00000836{"function":"runTask","parameter":{"id":174495930339100000,"name":"intermittent","actions":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174495932037200000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174495932037200000,"type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174495932037300000,"type":"arcOn","arcOnClass":0},{"segmentsNumber":2,"weldDistance":30,"intermittentSpeed":100,"interruptedWeldingOffId":174495932037300000,"id":174495932037300000,"type":"interruptedWeldingOn"},{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174495932037300000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174495932037300000,"type":"moveL"},{"interruptedWeldingOnId":174495932037300000,"id":174495932037300000,"type":"interruptedWeldingOff"},{"id":174495932037300000,"type":"arcOff"}],"type":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": 174495930339100000,
    "name": "intermitten",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 0,
              "y": 0,
              "z": 1176.5,
              "rx": 0,
              "ry": 0,
              "rz": 0,
              "J1": 0,
              "J2": 0,
              "J3": 0,
              "J4": 0,
              "J5": 0,
              "J6": 0,
              "TCP": "TCP",
              "UCS": "Base"
            },
            "enableVariable": 0,
            "variableSymbol": "p1",
            "id": 174495932037200000,
            "type": "point"
          }
        ],
        "speed": 100,
        "acceleration": 500,
        "radius": 10,
        "id": 174495932037200000,
        "type": "moveL"
      }
    ],
    "arcOn": {
      "arcOnMode": 0,
      "arcOnCurrent": 100,
      "arcOnAirPressure": 0,
      "arcOnVoltage": 0,
      "arcOnTime": 0,
    }
  }
}
```

```
"preflowTime": 0,
"arcOnPowerCorrection": 20,
"arcOnWireSpeed": 0,
"arcOnJobNumber": 1,
"arcOnInductance": 0,
"arcOnSlope": 0,
"arcOnArcControl": 0
},
"arcOff": {
  "arcOffMode": 0,
  "arcOffCurrent": 100,
  "arcOffVoltage": 0,
  "arcOffTime": 0,
  "arcOffDistance": 0,
  "enableArcOffDistance": true,
  "postflowTime": 0,
  "enableBurnback": 1,
  "burnBackTime": 0,
  "arcOffPowerCorrection": 20,
  "arcOffWireSpeed": 0,
  "arcOffJobNumber": 1,
  "arcOffInductance": 0,
  "arcOffSlope": 0,
  "arcOffArcControl": 0
},
"arcSet": {
  "arcSetMode": 0,
  "arcSetCurrent": 100,
  "arcSetAirPressure": 0,
  "arcSetVoltage": 0,
  "arcSetPowerCorrection": 20,
  "arcSetWireSpeed": 0,
  "arcSetJobNumber": 1,
  "arcSetInductance": 0,
  "arcSetScanSpeed": 200,
  "arcSetScanLength": 2,
  "arcSetSpduty": 50,
  "arcSetSpfreq": 2000,
  "arcSetSwingLength": 0,
  "arcSetSwingFrequency": 0,
  "speed": 10,
  "enablePushWire": 0,
  "arcSetSlope": 0,
  "arcSetArcControl": 0
},
```

```

    "weldProgram": {
      "isUseMemorySlot": 0,
      "programNumber": 1,
      "weldProcess": 0,
      "weldWireType": 0,
      "weldWireSize": 0,
      "weldGasType": 0
    },
    "id": 174495932037300000,
    "type": "arcOn",
    "arcOnClass": 0
  },
  {
    "segmentsNumber": 2,
    "weldDistance": 30,
    "intermittentSpeed": 100,
    "interruptedWeldingOffId": 174495932037300000,
    "id": 174495932037300000,
    "type": "interruptedWeldingOn"
  },
  {
    "children": [
      {
        "position": {
          "x": 0,
          "y": 0,
          "z": 1176.5,
          "rx": 0,
          "ry": 0,
          "rz": 0,
          "J1": 0,
          "J2": 0,
          "J3": 0,
          "J4": 0,
          "J5": 0,
          "J6": 0,
          "TCP": "TCP",
          "UCS": "Base"
        },
        "enableVariable": 0,
        "variableSymbol": "p1",
        "id": 174495932037300000,
        "type": "point"
      }
    ]
  },

```

```

    "speed": 100,
    "acceleration": 500,
    "radius": 10,
    "id": 174495932037300000,
    "type": "moveL"
  },
  {
    "interruptedWeldingOnId": 174495932037300000,
    "id": 174495932037300000,
    "type": "interruptedWeldingOff"
  },
  {
    "id": 174495932037300000,
    "type": "arcOff"
  }
],
"type": 0
}
}

```

表 4-2-17 间断焊模版-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: runTask
parameter	参数	object	见下表
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围: 0: 模拟任务, 不起弧; 1: 真实任务, 起弧;
actions	动作列表	array	见下表

actions[0]	直线		object	见 4.2.1.1 直线表
actions[1]	起弧		object	见 4.2.3.1 起弧表
actions[2]	间断焊 开始	id	string	动作 ID
		type	string	动作类型，固定值： interruptedWeldingOn
		segmentsNumber	int	分段数量
		weldDistance	int	焊接长度
		intermittentSpeed	int	空跑速度
children[3]	直线		object	见 4.2.1.1 直线表
children[4]	间断焊 结束	id	string	动作 ID
		type	string	固定值：interruptedWeldingOff
children[5]	收弧		object	见 4.2.3.2 收弧表

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.11 描述：电弧跟踪模版

发送消息：

```
####00000a6b{"function":"runTask","parameter":{"id":174495645270900000,"name":"zxc","actions":[{"children":[{"position":{"x":0,"y":0,"z":0,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174496126235300000,"type":"point"}],"speed":10,"acceleration":500,"radius":15,"id":174496126235300000,"type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnVoltageCompensation":0,"arcOnArcLength":0,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnInterpulse":0,"arcOnSlope":0,"arcOnArcControl":0,"arcOnProgramNumber":1},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffVoltageCompensation":0,"arcOffArcLength":0,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffInterpulse":0,"arcOffSlope":0,"arcOffArcControl":0,"arcOffProgramNumber":1,"pullWireTime":500},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":0,"arcSetVoltageCompensation":0,"arcSetArcLength":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetInterpulse":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174496126235300000,"type":"arcOn","arcOnClass":0},{"TrackMode":3,"YChangeRateThreashold":0.03,"YEARate":0.3,"YMaxD":20,"ZChangeRateThreashold":0.03,"ZEARate":0.1,"ZMaxD":5.5,"ZBaseCollectTime":30,"tcpMode":0,"eatrackOffid":174496126235400000,"id":174496126235300000,"type":"eatrackOn"},{"shape":0,"amplitude":2,"frequency":2,"dwellEnable":0,"dwellLeft":500,"dwellRight":500,"weaveFrameType":0,"centreRise":0,"elevation":0,"azimuth":0,"centralUplift":0,"suspendType":1,"id":174496126235300000,"type":"weaveOn"},{"children":[{"position":{"x":0,"y":0,"z":0,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","id":174496126235300000,"type":"point"}],"speed":10,"acceleration":500,"radius":15,"id":174496126235300000,"type":"moveL"},{"id":174496126235400000,"type":"weaveOff"},{"eatrackOnid":174496126235300000,"id":174496126235400000,"type":"eatrackOff"},{"id":174496126235400000,"type":"arcOff"}], "type":0}}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "runTask",
  "parameter": {
    "id": 174495645270900000,
    "name": "zxc",
    "actions": [
      {
        "children": [
```



```

    {
      "position": {
        "x": 0,
        "y": 0,
        "z": 0,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174496126235300000,
      "type": "point"
    }
  ],
  "speed": 10,
  "acceleration": 500,
  "radius": 15,
  "id": 174496126235300000,
  "type": "moveL"
},
{
  "arcOn": {
    "arcOnMode": 0,
    "arcOnCurrent": 100,
    "arcOnAirPressure": 0,
    "arcOnVoltage": 0,
    "arcOnVoltageCompensation": 0,
    "arcOnArcLength": 0,
    "arcOnTime": 0,
    "preflowTime": 0,
    "arcOnPowerCorrection": 20,
    "arcOnWireSpeed": 0,
    "arcOnJobNumber": 1,
    "arcOnInductance": 0,
    "arcOnInterpulse": 0,
    "arcOnSlope": 0,

```

```
"arcOnArcControl": 0,  
"arcOnProgramNumber": 1  
},  
"arcOff": {  
  "arcOffMode": 0,  
  "arcOffCurrent": 100,  
  "arcOffVoltage": 0,  
  "arcOffVoltageCompensation": 0,  
  "arcOffArcLength": 0,  
  "arcOffTime": 0,  
  "arcOffDistance": 0,  
  "enableArcOffDistance": true,  
  "postflowTime": 0,  
  "enableBurnback": 1,  
  "burnBackTime": 0,  
  "arcOffPowerCorrection": 20,  
  "arcOffWireSpeed": 0,  
  "arcOffJobNumber": 1,  
  "arcOffInductance": 0,  
  "arcOffInterpulse": 0,  
  "arcOffSlope": 0,  
  "arcOffArcControl": 0,  
  "arcOffProgramNumber": 1,  
  "pullWireTime": 500  
},  
"arcSet": {  
  "arcSetMode": 0,  
  "arcSetCurrent": 100,  
  "arcSetAirPressure": 0,  
  "arcSetVoltage": 0,  
  "arcSetVoltageCompensation": 0,  
  "arcSetArcLength": 0,  
  "arcSetPowerCorrection": 20,  
  "arcSetWireSpeed": 0,  
  "arcSetJobNumber": 1,  
  "arcSetInductance": 0,  
  "arcSetInterpulse": 0,  
  "arcSetScanSpeed": 200,  
  "arcSetScanLength": 2,  
  "arcSetSpduty": 50,  
  "arcSetSpfreq": 2000,  
  "arcSetSwingLength": 0,  
  "arcSetSwingFrequency": 0,  
  "speed": 10,  
  "enablePushWire": 0,
```

```
        "arcSetSlope": 0,
        "arcSetArcControl": 0
    },
    "weldProgram": {
        "isUseMemorySlot": 0,
        "programNumber": 1,
        "weldProcess": 0,
        "weldWireType": 0,
        "weldWireSize": 0,
        "weldGasType": 0
    },
    "id": 174496126235300000,
    "type": "arcOn",
    "arcOnClass": 0
},
{
    "TrackMode": 3,
    "YChangeRateThreashold": 0.03,
    "YERate": 0.3,
    "YMaxD": 20,
    "ZChangeRateThreashold": 0.03,
    "ZERate": 0.1,
    "ZMaxD": 5.5,
    "ZBaseCollectTime": 30,
    "tcpMode": 0,
    "eatrackOffId": 174496126235400000,
    "id": 174496126235300000,
    "type": "eatrackOn"
},
{
    "shape": 0,
    "ampitude": 2,
    "frequency": 2,
    "dwellEnable": 0,
    "dwellLeft": 500,
    "dwellRight": 500,
    "weaveFrameType": 0,
    "centreRise": 0,
    "elevation": 0,
    "azimuth": 0,
    "centralUplift": 0,
    "suspendType": 1,
    "id": 174496126235300000,
    "type": "weaveOn"
},
```

```

{
  "children": [
    {
      "position": {
        "x": 0,
        "y": 0,
        "z": 0,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174496126235300000,
      "type": "point"
    }
  ],
  "speed": 10,
  "acceleration": 500,
  "radius": 15,
  "id": 174496126235300000,
  "type": "moveL"
},
{
  "id": 174496126235400000,
  "type": "weaveOff"
},
{
  "eatrackOnId": 174496126235300000,
  "id": 174496126235400000,
  "type": "eatrackOff"
},
{
  "id": 174496126235400000,
  "type": "arcOff"
}
],

```

```

    "type": 0
  }
}

```

表 4-2-18 电弧焊模版-消息字段说明

参数	名称		数据类型	内容
id	任务 ID		string	任务 ID
name	任务名称		string	任务名称
type	任务类型		int	取值范围： 0：模拟任务，不起弧； 1：真实任务，起弧；
actions	动作列表		array	见下表
actions[0]	直线		object	见 4.2.1.1 直线表
actions[1]	起弧		object	见 4.2.3.1 起弧表
actions[2]	电弧跟踪开始	id	string	动作（action）ID
		type	string	固定值：eatrackOn
		TrackMode	int	固定值：0：不跟随 1：左右跟随 2：上下跟随 3：全跟随
		YChangeRateThreshold	int	左右跟随电流变化率门限
		YEARate	int	左右跟随纠偏速率
		YMaxD	int	左右跟随累计最大调整距离

		ZChangeRateThreshold	int	上下跟随电流变化率门限
		ZERate	int	上下跟随纠偏速率
		ZMaxD	int	上下跟随累计最大调整距离
		ZBaseCollectTime	int	上下跟随基准电流采集轮次
		tcpMode	int	0: TCP 的 Z+向上 1: TCP 的 Z+向下
children[3]	摆动开始		object	见 4.2.1.4 摆动
actions[4]	直线		object	见 4.2.1.1 直线表
children[5]	摆动结束		object	见 4.2.1.4 摆动
children[6]	电弧跟踪结束	id	string	动作 (action) ID
		type	string	固定值: eatrackOff
children[7]	收弧		object	见 4.2.3.2 收弧表

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.12 描述：相贯线模版

发送消息:

```
####000009cb{"function":"runTask","parameter":{"id":174853835834300000,"name":"xiangguanxian","actions":[{"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","weldTemplateId":"","datumPoint":"","id":174853853327700000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174853853327700000,"type":"moveL"},{"arcOn":{"arcOnMode":0,"arcOnCurrent":0,"arcOnAirPressure":0,"arcOnVoltage":-10,"arcOnTime":0,"preflowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":1,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnSlope":0,"arcOnArcControl":0,"arcOnProgramNumber":1,"arcOff":{"arcOffMode":0,"arcOffCurrent":0,"arcOffVoltage":-10,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postflowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":1,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0,"arcOffProgramNumber":1,"arcSet":{"arcSetMode":0,"arcSetCurrent":0,"arcSetAirPressure":0,"arcSetVoltage":-10,"arcSetPowerCorrection":20,"arcSetWireSpeed":1,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArcControl":0,"pulseFineTuning":[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0],"arcSetProgramNumber":1,"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"id":174853853327700000,"type":"arcOn","arcOnClass":0},"children":[{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","weldTemplateId":"","datumPoint":"","id":174853853327800000,"type":"point"},{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","weldTemplateId":"","datumPoint":"","id":174853853327800000,"type":"point"},{"position":{"x":0,"y":0,"z":1176.5,"rx":0,"ry":0,"rz":0,"J1":0,"J2":0,"J3":0,"J4":0,"J5":0,"J6":0,"TCP":"TCP","UCS":"Base"},"enableVariable":0,"variableSymbol":"p1","weldTemplateId":"","datumPoint":"","id":174853853327800000,"type":"point"}],"speed":100,"acceleration":500,"radius":10,"id":174853853327800000,"type":"moveIntersecting"},{"id":174853853327800000,"type":"arcOff"},"type":1}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": 174853835834300000,
    "name": "xiangguanxian",
    "actions": [
      {
        "children": [
          {
            "position": {
              "x": 0,
              "y": 0,
              "z": 1176.5,
```

```

        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174853853327700000,
    "type": "point"
}
],
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": 174853853327700000,
"type": "moveL"
},
{
    "arcOn": {
        "arcOnMode": 0,
        "arcOnCurrent": 0,
        "arcOnAirPressure": 0,
        "arcOnVoltage": -10,
        "arcOnTime": 0,
        "preflowTime": 0,
        "arcOnPowerCorrection": 20,
        "arcOnWireSpeed": 1,
        "arcOnJobNumber": 1,
        "arcOnInductance": 0,
        "arcOnSlope": 0,
        "arcOnArcControl": 0,
        "arcOnProgramNumber": 1
    },
    "arcOff": {
        "arcOffMode": 0,
        "arcOffCurrent": 0,
        "arcOffVoltage": -10,
        "arcOffTime": 0,
    }
}

```



```
"arcOffDistance": 0,
"enableArcOffDistance": true,
"postflowTime": 0,
"enableBurnback": 1,
"burnBackTime": 0,
"arcOffPowerCorrection": 20,
"arcOffWireSpeed": 1,
"arcOffJobNumber": 1,
"arcOffInductance": 0,
"arcOffSlope": 0,
"arcOffArcControl": 0,
"arcOffProgramNumber": 1
},
"arcSet": {
  "arcSetMode": 0,
  "arcSetCurrent": 0,
  "arcSetAirPressure": 0,
  "arcSetVoltage": -10,
  "arcSetPowerCorrection": 20,
  "arcSetWireSpeed": 1,
  "arcSetJobNumber": 1,
  "arcSetInductance": 0,
  "arcSetScanSpeed": 200,
  "arcSetScanLength": 2,
  "arcSetSpduty": 50,
  "arcSetSpfreq": 2000,
  "arcSetSwingLength": 0,
  "arcSetSwingFrequency": 0,
  "speed": 10,
  "enablePushWire": 0,
  "arcSetSlope": 0,
  "arcSetArcControl": 0,
  "pulseFineTuning": [
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0
  ],
}
```

```

    "arcSetProgramNumber": 1
  },
  "weldProgram": {
    "isUseMemorySlot": 0,
    "programNumber": 1,
    "weldProcess": 0,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  },
  "id": 174853853327700000,
  "type": "arcOn",
  "arcOnClass": 0
},
{
  "children": [
    {
      "position": {
        "x": 0,
        "y": 0,
        "z": 1176.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
      },
      "enableVariable": 0,
      "variableSymbol": "p1",
      "id": 174853853327800000,
      "type": "point"
    },
    {
      "position": {
        "x": 0,
        "y": 0,
        "z": 1176.5,
        "rx": 0,
        "ry": 0,

```

```

        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174853853327800000,
    "type": "point"
},
{
    "position": {
        "x": 0,
        "y": 0,
        "z": 1176.5,
        "rx": 0,
        "ry": 0,
        "rz": 0,
        "J1": 0,
        "J2": 0,
        "J3": 0,
        "J4": 0,
        "J5": 0,
        "J6": 0,
        "TCP": "TCP",
        "UCS": "Base"
    },
    "enableVariable": 0,
    "variableSymbol": "p1",
    "id": 174853853327800000,
    "type": "point"
}
},
"speed": 100,
"acceleration": 500,
"radius": 10,
"id": 174853853327800000,
"type": "moveIntersecting"
},
{

```

```

        "id": 174853853327800000,
        "type": "arcOff"
    }
],
"type": 1
}
}

```

表 4-2-19 相贯线模版-消息字段说明

参数	名称	数据类型	内容
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围： 0：模拟任务，不起弧； 1：真实任务，起弧；
actions	动作列表	array	见下表
actions[0]	直线	object	见 4.2.1.1 直线表
actions[1]	起弧	object	见 4.2.3.1 起弧表
actions[2]	相贯线	object	见 4.2.1.4 相贯线
actions[3]	摆动开始	object	见 4.2.1.4 摆动
actions[7]	收弧	object	见 4.2.3.2 收弧表

返回消息

调用成功返回：

```

{"code":0,"error":""}

```



```

"parameter": {
  "id": "177083067401400000",
  "name": "f2",
  "actions": [
    {
      "children": [
        {
          "position": {
            "x": -449.349,
            "y": 894.518,
            "z": -142.981,
            "rx": -55.387,
            "ry": -11.427,
            "rz": -173.951,
            "J1": -65.351,
            "J2": 40.052,
            "J3": -69.765,
            "J4": 3.743,
            "J5": -44.481,
            "J6": -3.6,
            "TCP": "TCP_1",
            "UCS": "Base"
          },
          "taskId": "177083067401400000",
          "id": "177083294114800003",
          "order": 1,
          "parentId": "177083294114800002",
          "device": "Hansrobot_Elfin_0",
          "type": "point",
          "isFirst": "1",
          "nextSiblingId": null
        }
      ],
      "speed": 20,
      "acceleration": 500,
      "radius": 10,
      "id": "177083294114800002",
      "type": "moveL"
    },
    {
      "arcOn": {
        "arcOnMode": 0,
        "arcOnCurrent": 105,
        "arcOnAirPressure": 0,
        "arcOnVoltage": 0,

```

```
"arcOnVoltageCompensation": 0,  
"arcOnArcLength": 0,  
"arcOnTime": 0,  
"preflowTime": 0,  
"arcOnPowerCorrection": 20,  
"arcOnWireSpeed": 1,  
"arcOnJobNumber": 1,  
"arcOnInductance": 0,  
"arcOnInterpulse": 0,  
"arcOnSlope": 0,  
"arcOnArcControl": 0,  
"arcOnProgramNumber": 1  
},  
"arcOff": {  
  "arcOffMode": 0,  
  "arcOffCurrent": 100,  
  "arcOffVoltage": 0,  
  "arcOffVoltageCompensation": 0,  
  "arcOffArcLength": 0,  
  "arcOffTime": 0,  
  "arcOffDistance": 0,  
  "enableArcOffDistance": true,  
  "postflowTime": 0,  
  "enableBurnback": 1,  
  "burnBackTime": 0,  
  "arcOffPowerCorrection": 20,  
  "arcOffWireSpeed": 1,  
  "arcOffJobNumber": 1,  
  "arcOffInductance": 0,  
  "arcOffInterpulse": 0,  
  "arcOffSlope": 0,  
  "arcOffArcControl": 0,  
  "arcOffProgramNumber": 1,  
  "pullWireTime": 500  
},  
"arcSet": {  
  "arcSetMode": 0,  
  "arcSetCurrent": 105,  
  "arcSetAirPressure": 0,  
  "arcSetVoltage": 0,  
  "arcSetVoltageCompensation": 0,  
  "arcSetArcLength": 0,  
  "arcSetPowerCorrection": 20,  
  "arcSetWireSpeed": 1,  
  "arcSetJobNumber": 1,
```

```

        "arcSetInductance": 0,
        "arcSetInterpulse": 0,
        "arcSetScanSpeed": 200,
        "arcSetScanLength": 2,
        "arcSetSpduty": 50,
        "arcSetSpfreq": 2000,
        "arcSetSwingLength": 0,
        "arcSetSwingFrequency": 0,
        "speed": 22,
        "enablePushWire": 0,
        "arcSetSlope": 0,
        "arcSetArcControl": 0,
        "pulseFineTuning": [
            0,
            0,
            0,
            0,
            0,
            0,
            0,
            0,
            0,
            0,
            0,
            0
        ],
        "arcSetProgramNumber": 1
    },
    "weldProgram": {
        "isUseMemorySlot": 0,
        "programNumber": 1,
        "weldProcess": 0,
        "weldWireType": 0,
        "weldWireSize": 0,
        "weldGasType": 0
    },
    "id": "177083294114800004",
    "type": "arcOn",
    "arcOnClass": 0
},
{
    "targetArcVoltage": 150,
    "maxSpeed": 5,
    "maxRatio": 3,
    "minRatio": -3,
    "id": "177083294114800005",

```



```

        "type": "arcVoltageTrackOn"
    },
    {
        "children": [
            {
                "position": {
                    "x": -449.349,
                    "y": 894.518,
                    "z": -142.981,
                    "rx": -55.387,
                    "ry": -11.427,
                    "rz": -173.951,
                    "J1": -65.351,
                    "J2": 40.052,
                    "J3": -69.765,
                    "J4": 3.743,
                    "J5": -44.481,
                    "J6": -3.6,
                    "TCP": "TCP_1",
                    "UCS": "Base"
                },
                "id": "177083294114800007",
                "type": "point"
            }
        ],
        "speed": 20,
        "acceleration": 500,
        "radius": 10,
        "id": "177083294114800006",
        "type": "moveL"
    },
    {
        "id": "177083294114900000",
        "type": "arcVoltageTrackOff"
    },
    {
        "id": "177083294114900001",
        "type": "arcOff"
    }
],
"type": 0
}

```

表 4-2-19 弧压跟踪模版-消息字段说明

参数	名称		数据类型	内容
id	任务 ID		string	任务 ID
name	任务名称		string	任务名称
type	任务类型		int	取值范围： 0：模拟任务，不起弧； 1：真实任务，起弧；
actions	动作列表		array	见下表
actions[0]	直线		object	见 4.2.1.1 直线表
actions[1]	起弧		object	见 4.2.3.1 起弧表
actions[2]	弧压跟踪开始	id	string	动作（action）ID
		type	string	固定值：arcVoltageTrackOn
		targetArcVoltage	int	目标弧压值，默认值：150
		maxSpeed	double	最大跟踪速度(单位:mm/s)，默认值： 5
		maxRatio	double	最大误差比，默认值：3
		minRatio	double	最小误差比，默认值：-3
actions[3]	直线		object	见 4.2.1.1 直线表
children[4]	弧压跟踪结束	id	string	动作（action）ID
		type	string	固定值：arcVoltageTrackOff

children[5]	收弧	object	见 4.2.3.2 收弧表
-------------	----	--------	---------------

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.2.3.14 描述：原地摆动模版

发送消息：

```
####00000f0a{"function":"runTask","parameter":{"id":"177702098820500000","name":"f2","actions":[{"remarks":"","parentType":null,"children":[{"remarks":"","parentType":"moveL","children":[],"disabled":0,"jobNumber":1,"enableVariable":0,"variableSymbol":"","weldTemplateId":1,"datumPoint":"","position":{"x":-320.797,"y":718.886,"z":-356.777,"rx":-34.443,"ry":-1.011,"rz":178.468,"J1":-66.532,"J2":51.823,"J3":-70.165,"J4":-0.141,"J5":-55.372,"J6":2.6,"TCP":"TCP_1","UCS":"Base"},"taskId":"177702098820500000","id":"177702099670100001","order":1,"parentId":"177702099670100000","device":"Hansrobot_Elfin_0","type":"point","isFirst":"1","nextSiblingId":null}],disabled":0,"speed":100,"acceleration":500,"radius":10,"taskId":"177702098820500000","id":"177702099670100000","order":1,"parentId":null,"device":"Hansrobot_Elfin_0","type":"moveL","isFirst":"1","nextSiblingId":"177702099670100002"}],"remarks":"","parentType":null,"children":[],"disabled":0,"craftMode":2,"craftJob":null,"thickness":2,"arcOn":{"arcOnMode":0,"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnVoltage":0,"arcOnVoltageCompensation":0,"arcOnArcLength":0,"arcOnTime":200,"preFlowTime":0,"arcOnPowerCorrection":20,"arcOnWireSpeed":0,"arcOnJobNumber":1,"arcOnInductance":0,"arcOnInterpulse":0,"arcOnSlope":0,"arcOnArcControl":0,"arcOnProgramNumber":1},"arcOff":{"arcOffMode":0,"arcOffCurrent":100,"arcOffVoltage":0,"arcOffVoltageCompensation":0,"arcOffArcLength":0,"arcOffTime":0,"arcOffDistance":0,"enableArcOffDistance":true,"postFlowTime":0,"enableBurnback":1,"burnBackTime":0,"arcOffPowerCorrection":20,"arcOffWireSpeed":0,"arcOffJobNumber":1,"arcOffInductance":0,"arcOffInterpulse":0,"arcOffSlope":0,"arcOffArcControl":0,"arcOffProgramNumber":1,"pullWireTime":500},"arcSet":{"arcSetMode":0,"arcSetCurrent":100,"arcSetAirPressure":0,"arcSetVoltage":0,"arcSetVoltageCompensation":0,"arcSetArcLength":0,"arcSetPowerCorrection":20,"arcSetWireSpeed":0,"arcSetJobNumber":1,"arcSetInductance":0,"arcSetInterpulse":0,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingLength":0,"arcSetSwingFrequency":0,"speed":10,"enablePushWire":0,"arcSetSlope":0,"arcSetArc
```

```
Control":0,"scanParams":{"pattern":0,"param1":1,"param2":1,"rotationAngle":0,"frequency":
50,"pullbackLength":0,"addWireLength":0},"pulseFineTuning":[0,0,0,0,0,0,0,0,0,0],"arcSe
tProgramNumber":1,"arcSetUnifyMode":0},"weldProgram":{"isUseMemorySlot":0,"program
Number":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0},"taskId":
"177702098820500000","id":"177702099670100002","order":2,"parentId":null,"device":
"Megmeet_350_3","type":"arcOn","isFirst":0,"nextSiblingId":"177702099670200000","arc
OnClass":0},{ "remarks":"","parentType":null,"disabled":0,"amplitude":2,"frequency":2,"dwell
Enable":0,"dwellLeft":500,"dwellRight":500,"logic":{"type":"readlo","expression":[{"operand
":"CIO","operator":"=","value":1}]},"taskId":"177702098820500000","id":"17770209967020
0000","order":3,"parentId":null,"device":"Megmeet_350_3","type":"weaveInPlace","isFirst":
"0","nextSiblingId":"177702099670200002","children":[{"remarks":"","parentType":"weaveI
nPlace","disabled":0,"jobNumber":1,"enableVariable":0,"variableSymbol":"","weldTemplatel
d":1,"datumPoint":"","position":{"x":-320.797,"y":624.011,"z":-356.777,"rx":-34.443,"ry":-1.
011,"rz":178.468,"J1":-63.4,"J2":42.839,"J3":-86.626,"J4":-0.351,"J5":-47.906,"J6":5.883,
"TCP":"TCP_1","UCS":"Base"},"taskId":"177702098820500000","id":"1777020996702000
01","order":1,"parentId":"177702099670200000","device":"Hansrobot_Elfin_0","type":"poi
nt","isFirst":1,"nextSiblingId":null,"children":[]}]},"remarks":"","parentType":null,"children":
[],"disabled":0,"taskId":"177702098820500000","id":"177702099670200002","order":4,"p
arentId":null,"device":"Megmeet_350_3","type":"arcOff","isFirst":0,"nextSiblingId":null},"t
ype":0,"enableRealtimeMod":0}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "runTask",
  "parameter": {
    "id": "177702098820500000",
    "name": "f2",
    "actions": [
      {
        "remarks": "",
        "parentType": null,
        "children": [
          {
            "remarks": "",
            "parentType": "moveL",
            "children": [

            ],
            "disabled": 0,
            "jobNumber": 1,
            "enableVariable": 0,
            "variableSymbol": "",
            "weldTemplateId": 1,
            "datumPoint": "",
```

```

    "position": {
      "x": -320.797,
      "y": 718.886,
      "z": -356.777,
      "rx": -34.443,
      "ry": -1.011,
      "rz": 178.468,
      "J1": -66.532,
      "J2": 51.823,
      "J3": -70.165,
      "J4": -0.141,
      "J5": -55.372,
      "J6": 2.6,
      "TCP": "TCP_1",
      "UCS": "Base"
    },
    "taskId": "177702098820500000",
    "id": "177702099670100001",
    "order": 1,
    "parentId": "177702099670100000",
    "device": "Hansrobot_Elfin_0",
    "type": "point",
    "isFirst": "1",
    "nextSiblingId": null
  }
],
"disabled": 0,
"speed": 100,
"acceleration": 500,
"radius": 10,
"taskId": "177702098820500000",
"id": "177702099670100000",
"order": 1,
"parentId": null,
"device": "Hansrobot_Elfin_0",
"type": "moveL",
"isFirst": "1",
"nextSiblingId": "177702099670100002"
},
{
  "remarks": "",
  "parentType": null,
  "children": [

],

```

```
"disabled": 0,  
"craftMode": 2,  
"craftJob": null,  
"thickness": 2,  
"arcOn": {  
  "arcOnMode": 0,  
  "arcOnCurrent": 100,  
  "arcOnAirPressure": 0,  
  "arcOnVoltage": 0,  
  "arcOnVoltageCompensation": 0,  
  "arcOnArcLength": 0,  
  "arcOnTime": 200,  
  "preflowTime": 0,  
  "arcOnPowerCorrection": 20,  
  "arcOnWireSpeed": 0,  
  "arcOnJobNumber": 1,  
  "arcOnInductance": 0,  
  "arcOnInterpulse": 0,  
  "arcOnSlope": 0,  
  "arcOnArcControl": 0,  
  "arcOnProgramNumber": 1  
},  
"arcOff": {  
  "arcOffMode": 0,  
  "arcOffCurrent": 100,  
  "arcOffVoltage": 0,  
  "arcOffVoltageCompensation": 0,  
  "arcOffArcLength": 0,  
  "arcOffTime": 0,  
  "arcOffDistance": 0,  
  "enableArcOffDistance": true,  
  "postflowTime": 0,  
  "enableBurnback": 1,  
  "burnBackTime": 0,  
  "arcOffPowerCorrection": 20,  
  "arcOffWireSpeed": 0,  
  "arcOffJobNumber": 1,  
  "arcOffInductance": 0,  
  "arcOffInterpulse": 0,  
  "arcOffSlope": 0,  
  "arcOffArcControl": 0,  
  "arcOffProgramNumber": 1,  
  "pullWireTime": 500  
},  
"arcSet": {
```

```
"arcSetMode": 0,  
"arcSetCurrent": 100,  
"arcSetAirPressure": 0,  
"arcSetVoltage": 0,  
"arcSetVoltageCompensation": 0,  
"arcSetArcLength": 0,  
"arcSetPowerCorrection": 20,  
"arcSetWireSpeed": 0,  
"arcSetJobNumber": 1,  
"arcSetInductance": 0,  
"arcSetInterpulse": 0,  
"arcSetScanSpeed": 200,  
"arcSetScanLength": 2,  
"arcSetSpduty": 50,  
"arcSetSpfreq": 2000,  
"arcSetSwingLength": 0,  
"arcSetSwingFrequency": 0,  
"speed": 10,  
"enablePushWire": 0,  
"arcSetSlope": 0,  
"arcSetArcControl": 0,  
"scanParams": {  
  "pattern": 0,  
  "param1": 1,  
  "param2": 1,  
  "rotationAngle": 0,  
  "frequency": 50,  
  "pullbackLength": 0,  
  "addWireLength": 0  
},  
"pulseFineTuning": [  
  0,  
  0,  
  0,  
  0,  
  0,  
  0,  
  0,  
  0,  
  0,  
  0,  
  0,  
  0  
],  
"arcSetProgramNumber": 1,  
"arcSetUnifyMode": 0
```

```

    },
    "weldProgram": {
      "isUseMemorySlot": 0,
      "programNumber": 1,
      "weldProcess": 0,
      "weldWireType": 0,
      "weldWireSize": 0,
      "weldGasType": 0
    },
    "taskId": "177702098820500000",
    "id": "177702099670100002",
    "order": 2,
    "parentId": null,
    "device": "Megmeet_350_3",
    "type": "arcOn",
    "isFirst": "0",
    "nextSiblingId": "177702099670200000",
    "arcOnClass": 0
  },
  {
    "remarks": "",
    "parentType": null,
    "disabled": 0,
    "amplitude": 2,
    "frequency": 2,
    "dwellEnable": 0,
    "dwellLeft": 500,
    "dwellRight": 500,
    "logic": {
      "type": "readlo",
      "expression": [
        {
          "operand": "CI0",
          "operator": "=",
          "value": 1
        }
      ]
    },
    "taskId": "177702098820500000",
    "id": "177702099670200000",
    "order": 3,
    "parentId": null,
    "device": "Megmeet_350_3",
    "type": "weaveInPlace",
    "isFirst": "0",

```



```

    "nextSiblingId": "177702099670200002",
    "children": [
      {
        "remarks": "",
        "parentType": "weaveInPlace",
        "disabled": 0,
        "jobNumber": 1,
        "enableVariable": 0,
        "variableSymbol": "",
        "weldTemplateId": 1,
        "datumPoint": "",
        "position": {
          "x": -320.797,
          "y": 624.011,
          "z": -356.777,
          "rx": -34.443,
          "ry": -1.011,
          "rz": 178.468,
          "J1": -63.4,
          "J2": 42.839,
          "J3": -86.626,
          "J4": -0.351,
          "J5": -47.906,
          "J6": 5.883,
          "TCP": "TCP_1",
          "UCS": "Base"
        },
        "taskId": "177702098820500000",
        "id": "177702099670200001",
        "order": 1,
        "parentId": "177702099670200000",
        "device": "Hansrobot_Elfin_0",
        "type": "point",
        "isFirst": "1",
        "nextSiblingId": null,
        "children": [

        ]
      }
    ]
  },
  {
    "remarks": "",
    "parentType": null,
    "children": [

```

```

    ],
    "disabled": 0,
    "taskId": "177702098820500000",
    "id": "177702099670200002",
    "order": 4,
    "parentId": null,
    "device": "Megmeet_350_3",
    "type": "arcOff",
    "isFirst": "0",
    "nextSiblingId": null
  }
],
"type": 0,
"enableRealtimeMod": 0
}
}

```

表 4-2-19 弧压跟踪模版-消息字段说明

参数	名称	数据类型	内容
id	任务 ID	string	任务 ID
name	任务名称	string	任务名称
type	任务类型	int	取值范围： 0：模拟任务，不起弧； 1：真实任务，起弧；
enableRealtimeMod	预留	int	默认填 0
actions	动作列表	array	见下表
actions[0]	直线	object	见 4.2.1.1 直线表
actions[1]	起弧	object	见 4.2.3.1 起弧表

actions[2]	原地摆动	id	string	动作 (action) ID
		type	string	固定值: weaveInPlace
		amplitude	double	振幅
		frequency	double	频率
		maxRatio	double	最大误差比, 默认值: 3
		dwellEnable	int	是否两端停留, 固定值: 0: 否 1: 是
		dwellLeft	double	左端停留, 停留时间, 单位 ms, 端侧停留有效, 推荐 100ms 以上
		dwellRight	double	右端停留, 停留时间, 单位 ms, 端侧停留有效, 推荐 100ms 以上
		logic	object	读取 IO 信号(CI*/DI*)高低: <pre>{ "logic": { "type": "readIo", "expression": [{ "operand": "CI0", "operator": "=", "value": 1}] } }</pre>

				<pre>"operand": "timeout", "operator": "=", "value": 1000}] }</pre>
children[3]	收弧		object	见 4.2.3.2 收弧表

返回消息

调用成功返回：

```
{"code":0,"error":""}
```

调用失败返回：

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.3 焊接任务控制指令

4.3.1 PauseTask

4.3.1.1 描述：暂停任务

发送消息：

```
####00000027{"function":"pauseTask","parameter":""}$$$$
```

消息体 JSON 格式化：

```
{
  "function": "pauseTask",
  "parameter": ""
}
```

表 4-3-1 暂停任务-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: pauseTask
parameter	参数	string	空字符串

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.3.2 ContinueTask

4.3.2.1 描述：继续任务

发送消息:

```
####0000006a{"function":"continueTask","parameter":{"continueType":0,"backtrackingDistance":5,"backtrackingSpeed":50}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "continueTask",
  "parameter": {
    "continueType": 0,
    "backtrackingDistance": 5,
    "backtrackingSpeed": 50
  }
}
```

表 4-3-2 继续任务-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: continueTask
parameter	参数	object	见下表
continueType	继续类型	int	0: 原始路径恢复 1: 重新再启动
backtrackingDistance	回退距离	int	continueType 为 0 时生效。暂停位置回退路径上的距离。
backtrackingSpeed	回退速度	int	continueType 为 0 时生效。回退时的速度。

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.3.3 StopTask

4.3.3.1 描述: 停止任务

发送消息:

```
####00000026{"function":"stopTask","parameter":""}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "stopTask",
  "parameter": ""
}
```

表 4-3-3 停止任务-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: stopTask
parameter	参数	string	空字符串

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.3.4 ReservationStatus

4.3.4.1 描述: 预约任务状态

发送消息:

```
####00000030{"function":"reservationStatus","parameter":"1"}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "reservationStatus",
  "parameter": 1
}
```

表 4-3-4 预约任务-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: setAotaiMode
parameter	参数	int	0:关闭预约, 1:开启预约

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.3.5 Real-time welding parameter modification

4.3.5.1 描述: 实时焊接参数修改

发送消息:

```
####00000047{"function":"modifyRuntimeParam","parameter":{"paramType":1,"value":6}}
$$$$
```

消息体 JSON 格式化:

```
{
  "function": "modifyRuntimeParam",
  "parameter": {
    "paramType": 1,
    "value": 6
  }
}
```

表 4-3-4 预约任务-消息字段说明

参数	名称	数据类型	内容
----	----	------	----

function	功能名称	string	固定值: modifyRuntimeParam
parameter	参数	object	见下表
paramType	参数类型	int	0: 代表电流; 1: 电压; 2: 送丝速度
value	参数值	double	参数类型值, 需要下发给焊机的实际值

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

4.4 工艺库

4.4.1 ChangeArcSet

4.4.1.1 描述: 设置焊接参数

发送消息:

```
####00000409{"function":"changeArcSet","parameter":{"arcOff":{"arcOffCurrent":100,"arcOffJobNumber":0,"arcOffMode":0,"arcOffPowerCorrection":0,"arcOffTime":100,"arcOffDistance":0,"enableArcOffDistance":false,"arcOffVoltage":50,"arcOffWireSpeed":20,"burnBackTime":0,"enableBurnback":1,"postflowTime":200,"arcOffInductance":0,"arcOffSlope":0,"arcOffArcControl":0},"arcSet":{"arcSetCurrent":100,"arcSetJobNumber":0,"arcSetMode":1,"arcSetPowerCorrection":0,"arcSetVoltage":50,"arcSetWireSpeed":20,"arcSetInductance":0,"arcSetSlope":0,"speed":10,"arcSetScanSpeed":200,"arcSetScanLength":2,"arcSetSpduty":50,"arcSetSpfreq":2000,"arcSetSwingFrequency":0,"arcSetSwingLength":0,"arcSetArcControl":0},"arcOn":{"arcOnCurrent":100,"arcOnAirPressure":0,"arcOnJobNumber":0,"arcOn
```

```
Mode":1,"arcOnPowerCorrection":0,"arcOnTime":200,"arcOnVoltage":30,"arcOnWireSpeed":20,"arcOnInductance":0,"arcOnSlope":0,"preflowTime":0,"arcOnArcControl":0},"weldProgram":{"isUseMemorySlot":0,"programNumber":1,"weldProcess":0,"weldWireType":0,"weldWireSize":0,"weldGasType":0}}}}$$$$
```

消息体 JSON 格式化:

```
{
  "function": "changeArcSet",
  "parameter": {
    "arcOff": {
      "arcOffCurrent": 100,
      "arcOffJobNumber": 0,
      "arcOffMode": 0,
      "arcOffPowerCorrection": 0,
      "arcOffTime": 100,
      "arcOffDistance": 0,
      "enableArcOffDistance": false,
      "arcOffVoltage": 50,
      "arcOffWireSpeed": 20,
      "burnBackTime": 0,
      "enableBurnback": 1,
      "postflowTime": 200,
      "arcOffInductance": 0,
      "arcOffSlope": 0,
      "arcOffArcControl": 0
    },
    "arcSet": {
      "arcSetCurrent": 100,
      "arcSetJobNumber": 0,
      "arcSetMode": 1,
      "arcSetPowerCorrection": 0,
      "arcSetVoltage": 50,
      "arcSetWireSpeed": 20,
      "arcSetInductance": 0,
      "arcSetSlope": 0,
      "speed": 10,
      "arcSetScanSpeed": 200,
      "arcSetScanLength": 2,
      "arcSetSpduty": 50,
      "arcSetSpfreq": 2000,
      "arcSetSwingFrequency": 0,
      "arcSetSwingLength": 0,
      "arcSetArcControl": 0
    }
  },
}
```

```

"arcOn": {
  "arcOnCurrent": 100,
  "arcOnAirPressure": 0,
  "arcOnJobNumber": 0,
  "arcOnMode": 1,
  "arcOnPowerCorrection": 0,
  "arcOnTime": 200,
  "arcOnVoltage": 30,
  "arcOnWireSpeed": 20,
  "arcOnInductance": 0,
  "arcOnSlope": 0,
  "preflowTime": 0,
  "arcOnArcControl": 0
},
"weldProgram": {
  "isUseMemorySlot": 0,
  "programNumber": 1,
  "weldProcess": 0,
  "weldWireType": 0,
  "weldWireSize": 0,
  "weldGasType": 0
}
}
}

```

表 4-4-1 设置焊接参数-消息字段说明

参数	名称	数据类型	内容
function	功能名称	string	固定值: changeArcSet
parameter	参数	object	见下表
type	是否启用	string	固定值: arcSet
arcOff	收弧	object	见 4.1.7
arcOn	起弧	object	见 4.1.6
arcSet	下发电流电压	object	见 4.1.5

weldProgram	焊接程序	object	见 4.1.20 (仅米勒 ModBus TCP/IP 模式)
-------------	------	--------	---------------------------------

返回消息

调用成功返回:

```
{"code":0,"error":""}
```

调用失败返回:

```
{"code":102101,"error":"error message"}
```

具体错误码参见 附录 1

第五章 焊机设备取值

表 5-1 焊机通信协议类型及对应数值

协议取值	协议描述
0	IO 信号/模拟量
2	ModbusTCP
3	Eethernet/IP
5	RS485

5.1 设备名称/协议

表 5-2 焊接设备品牌与通信协议对照表

品牌	数据类型	内容	对接协议
麦格米特	string/int	"Megmeet"	0,3

克鲁斯	string/int	"Cloos"	0,3
米勒	string/int	"Miller"	0,2,3
福尼斯	string/int	"Fronius"	0,2
松下	string/int	"Panasonic"	0,3
林肯	string/int	"Lincoln"	0,3
福乐玛	string/int	"Flama"	0,3
威尔泰克	string/int	"WeldTEC"	0,2,3
肯比	string/int	"Kemppi"	0,2
宾采尔	string/int	"Binzel"	0,2
奥太	string/int	"Aotai"	0,2
洛驰	string/int	"Lorch"	0,2
伊萨	string/int	"ESAB"	0,3
默克莱	string/int	"Merkle"	0,3
伊达新	string/int	"EWM"	0,3
海宝	string/int	"Hypertherm"	5
瑞凌	string/int	"Riland"	0

5.2 设备型号

表 5-2 焊接设备型号对照表

品牌	数据类型	模式
Megmeet	string	"350","500","630"
Binzel	string	"500"
Aotai	string	"500","630"
Fronius	string	"500"
Panasonic	string	"G","FT"
EWM	string	"EWM"
Flama	string	"250","350","500","1000","1000_RS485"
WeldTEC	string	"DP/DPL","NB"
Kemppi	string	"500"
Cloos	string	"Cloos"
Merkle	string	"352","452","552"
Lorch	string	"Lorch"
Lincoln	string	"Lincoln","R450"
ESAB	string	"ESAB"
Miller	string	"Miller"

Riland	string	"Riland"
Hypertherm	string	"65","85","105"

5.3 焊接模式

表 5-2 焊接设备支持模式及编号说明

品牌	型号/协议	模式
Megmeet	无	0: 直流一元化 1: 脉冲一元化 2: JOB 模式 3: 断续焊 4: 直流分别化
Binzel	无	0: 平特性, 1: 脉冲程序 2: JOB 模式 3: 低飞溅
Aotai	无	0: 平特性 1: 脉冲程序 2: JOB 模式 3: 低飞溅
Fronius	无	1: 特殊的两步法模式特性 2: JOB 模式 3: 两步模式特性 4: 两步 MIG/MAG 标准手册
Panasonic	无	0: 独立调节模式 2: 一元化调节模式 4: JOB 模式

EWM	无	0: 常规焊模式 1: 脉冲焊模式
Flama	welderProtocol = 2 或 welderProtocol = 5	0: 直流分别化 1: 直流一元化 3: 脉冲一元化 4: JOB 模式
Flama	welderProtocol = 3	0: 直流一元化 1: 脉冲一元化 2: JOB 模式 3: 近控模式
WeldTEC	welderModel = 'DP/DPL'	0: 面板控制 1: 调用模式 2: 恒压 3: 单脉冲 4: 双脉冲
WeldTEC	welderModel = 'NB'	0: 脉冲一元化 1: 脉冲分别化 2: 恒压一元化 3: 恒压分别化
Kemppi	无	0: JOB 模式
Cloos	无	0: JOB 模式
Merkle	无	0: 脉冲 1: MIG/MAG 5: JOB 模式 6: 特殊脉冲(快速) 7: 特殊脉冲(慢速)

Lorch	无	7: JOB 模式
Lincoln	无	1: JOB 模式
ESAB	无	0: JOB 模式
Miller	无	2: JOB 模式

第六章 接口使用示例

6.1 python 运行任务

```
import socket
import json

def main():
    # 定义服务器地址和端口
    host = '127.0.0.1'
    port = 30600
    task = {
        "id": 174495645270900000,
        "name": "zxc",
        "actions": [
            {
                "children": [
                    {
                        "position": {
                            "x": 0,
                            "y": 0,
                            "z": 0,
                            "rx": 0,
                            "ry": 0,
                            "rz": 0,
                            "J1": 0,
                            "J2": 0,
                            "J3": 0,
                            "J4": 0,
                            "J5": 0,
                            "J6": 0,
                            "TCP": "TCP",
                            "UCS": "Base"
                        },
                        "enableVariable": 0,
                        "variableSymbol": "p1",
                        "id": 174496126235300000,
                        "type": "point"
                    }
                ],
                "speed": 10,
                "acceleration": 500,
```

```
"radius": 15,
"id": 174496126235300000,
"type": "moveL"
},
{
  "arcOn": {
    "arcOnMode": 0,
    "arcOnCurrent": 100,
    "arcOnAirPressure": 0,
    "arcOnVoltage": 0,
    "arcOnTime": 0,
    "preflowTime": 0,
    "arcOnPowerCorrection": 20,
    "arcOnWireSpeed": 0,
    "arcOnJobNumber": 1,
    "arcOnInductance": 0,
    "arcOnSlope": 0,
    "arcOnArcControl": 0
  },
  "arcOff": {
    "arcOffMode": 0,
    "arcOffCurrent": 100,
    "arcOffVoltage": 0,
    "arcOffTime": 0,
    "arcOffDistance": 0,
    "enableArcOffDistance": true,
    "postflowTime": 0,
    "enableBurnback": 1,
    "burnBackTime": 0,
    "arcOffPowerCorrection": 20,
    "arcOffWireSpeed": 0,
    "arcOffJobNumber": 1,
    "arcOffInductance": 0,
    "arcOffSlope": 0,
    "arcOffArcControl": 0
  },
  "arcSet": {
    "arcSetMode": 0,
    "arcSetCurrent": 100,
    "arcSetAirPressure": 0,
    "arcSetVoltage": 0,
    "arcSetPowerCorrection": 20,
    "arcSetWireSpeed": 0,
    "arcSetJobNumber": 1,
    "arcSetInductance": 0,
```

```

    "arcSetScanSpeed": 200,
    "arcSetScanLength": 2,
    "arcSetSpduty": 50,
    "arcSetSpfreq": 2000,
    "arcSetSwingLength": 0,
    "arcSetSwingFrequency": 0,
    "speed": 10,
    "enablePushWire": 0,
    "arcSetSlope": 0,
    "arcSetArcControl": 0
  },
  "weldProgram": {
    "isUseMemorySlot": 0,
    "programNumber": 1,
    "weldProcess": 0,
    "weldWireType": 0,
    "weldWireSize": 0,
    "weldGasType": 0
  },
  id": 174496126235300000,
  "type": "arcOn",
  "arcOnClass": 0
},
{
  "TrackMode": 3,
  "YChangeRateThreashold": 0.03,
  "YEARate": 0.3,
  "YMaxD": 20,
  "ZChangeRateThreashold": 0.03,
  "ZEARate": 0.1,
  "ZMaxD": 5.5,
  "ZBaseCollectTime": 30,
  "eatrackOffId": 174496126235400000,
  id": 174496126235300000,
  "type": "eatrackOn"
},
{
  "shape": "zigzag",
  "ampitude": 2,
  "frequency": 2,
  "dwellEnable": 0,
  "dwellLeft": 500,
  "dwellRight": 500,
  "weaveFrameType": 0,
  "centreRise": 0,

```

```

    "elevation": 0,
    "azimuth": 0,
    "id": 174496126235300000,
    "type": "weaveOn"
  },
  {
    "children": [
      {
        "position": {
          "x": 0,
          "y": 0,
          "z": 0,
          "rx": 0,
          "ry": 0,
          "rz": 0,
          "J1": 0,
          "J2": 0,
          "J3": 0,
          "J4": 0,
          "J5": 0,
          "J6": 0,
          "TCP": "TCP",
          "UCS": "Base"
        },
        "enableVariable": 0,
        "variableSymbol": "p1",
        "id": 174496126235300000,
        "type": "point"
      }
    ],
    "speed": 10,
    "acceleration": 500,
    "radius": 15,
    "id": 174496126235300000,
    "type": "moveL"
  },
  {
    "id": 174496126235400000,
    "type": "weaveOff"
  },
  {
    "eatrackOnId": 174496126235300000,
    "id": 174496126235400000,
    "type": "eatrackOff"
  },

```

```
{
    {
        id": 174496126235400000,
        "type": "arcOff"
    }
},
"type": 0
}
}
```

```
send_start_flag = '####'
send_end_flag = '$$$$'

def get_byte_length(content):
    return len(content.encode('utf-8'))

content = {
    "function": "runTask",
    "parameter": task
}

send_content = json.dumps(content)
byte_length = hex(get_byte_length(send_content))[2:].zfill(8)
send_content = send_start_flag + byte_length + send_content + send_end_flag

# 创建一个 socket 对象
with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
    # 连接到服务器
    s.connect((host, port))

    # 发送数据
    s.sendall(send_content.encode('utf-8'))

    # 接收数据
    data = s.recv(1024) # 接收 1024 字节的数据

    # 打印接收到的信息
    print('Received', data.decode('utf-8'))

if __name__ == '__main__':
    main()
```

附录 1：错误码说明

序号	返回值	名称	数据类型	内容
公共				
1	1	错误码	int	命令错误
任务				
2	101001	错误码	int	该任务不支持或者禁止运行
3	101002	错误码	int	任务计算路径错误
4	101003	错误码	int	任务参数设置错误
5	101004	错误码	int	任务的函数名称错误
6	101005	错误码	int	存在未设置值的参数
7	101006	错误码	int	任务内部计算错误
8	101007	错误码	int	点位设置错误
9	101008	错误码	int	命令错误
10	101009	错误码	int	设定参数不合理：amplitude, intervalDistance < 0.000001 摆焊每周期的运动弧长距离
11	101010	错误码	int	weaveFrameType = 0 时，工具坐标系的 Z 方向和焊道方向同向
12	101011	错误码	int	圆弧参数点位不共面

13	101012	错误码	int	间断焊起弧点要在间断焊开启前
14	101013	错误码	int	逻辑判断条件不满足
15	101014	错误码	int	用户未确认视觉轨迹
16	101015	错误码	int	不支持该继续类型
设备				
17	102001	错误码	int	未知的设备类型
18	102101	错误码	int	焊机连接失败
19	102102	错误码	int	焊机响应超时
20	102103	错误码	int	焊机校验失败
21	102104	错误码	int	向焊机发送数据失败
22	102105	错误码	int	从焊机接收数据失败
23	102106	错误码	int	焊机起弧失败
24	102107	错误码	int	焊机收弧失败
25	102108	错误码	int	焊机设备忙碌中
26	102109	错误码	int	焊机协议不支持
27	102110	错误码	int	焊机电流异常

28	102111	错误码	int	焊机不支持该功能
29	102112	错误码	int	触碰寻位失败
30	102113	错误码	int	弧压跟踪电流过小
31	102114	错误码	int	焊接未连接
32	102115	错误码	int	焊接断开连接
激光				
33	102201	错误码	int	激光连接失败
34	102202	错误码	int	激光打开失败
35	102203	错误码	int	向激光发送数据失败
36	102204	错误码	int	从激光接收数据失败
37	102205	错误码	int	从激光请求数据失败
38	102206	错误码	int	激光寻位失败
39	102207	错误码	int	校准初始化失败
40	102208	错误码	int	校准点位失败
41	102209	错误码	int	激光跟踪失败
42	102210	错误码	int	激光跟踪返回失败

43	102211	错误码	int	激光寻道失败
44	102212	错误码	int	激光传感器未打开
相机				
45	102301	错误码	int	未知的相机名称
46	102302	错误码	int	相机未连接
47	102303	错误码	int	相机连接失败
48	102304	错误码	int	相机启动失败
49	102305	错误码	int	向相机发送数据错误
50	102306	错误码	int	从相机接收数据失败
51	102307	错误码	int	从相机接收数据超过限制
52	102308	错误码	int	相机命令执行失败
53	102309	错误码	int	相机拍照失败
IO				
54	102401	错误码	int	IO 信号异常
切割				
55	102501	错误码	int	切割机协议错误

机器人本体及控制器				
56	103001	错误码	int	下发点位失败
57	103002	错误码	int	机器人响应失败
58	103003	错误码	int	机器人当前状态不允许执行该操作
59	103004	错误码	int	算法计算返回错误
程序主控				
60	104001	错误码	int	创建 socket 错误
61	104002	错误码	int	socket 绑定错误
62	104003	错误码	int	socket 监听错误
63	104004	错误码	int	创建多路复用错误
64	104005	错误码	int	多路复用控制错误
65	104006	错误码	int	websocket 通讯异常，建议重启服务